

RESEARCH REPORT

Allocating AI Translation Compute for Marginal Educational Access

A global language, interlanguage, accessibility, and open-curriculum
portfolio model

A global portfolio model for language, shared-core, accessibility, and open-curriculum decisions

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Abstract

Artificial intelligence sharply reduces some costs of adapting open educational resources, but “translate into the largest language” is not an adequate allocation rule. A useful portfolio must ask how many people gain *new, comfortable, usable* access; which content is absent; whether another edition or academic lingua franca already serves the same readers; whether the output works in the relevant script, device, bandwidth, and disability context; and how much source preparation, terminology work, mathematical checking, correction, and maintenance the edition requires. This paper constructs a reproducible global allocation model around that marginal-access question. It joins dated language-population observations to exact language/variety/script/territory interventions, local Open Logic and OpenStax workloads, advanced-material scarcity proxies, accessibility requirements, and evidence-bounded interlanguage architectures. The common language-priority comparator is a 210-unit, 120,083-source-token formal-reasoning curriculum; larger OpenStax portfolios are costed separately so that useful breadth is not mistaken for inefficiency. A population-specific needs layer then separates foundational numeracy recovery, bilingual academic-register transfer, secondary-to-tertiary bridging, coherent undergraduate sequences, and advanced/reference work: population rank does not select the same textbook for everyone. Gross demographic ceilings, evidence-backed marginal floors, and wide neutral sensitivities are reported separately. Constructed bridges receive no blanket family reach; shared semantic cores receive compute savings only through explicitly named localized outputs. The corrected source register contains 475 population observations and 81 registered source records from 58 distinct authority labels; 143 exact natural-language/profile interventions were scored and 134 passed the cardinal eligibility gate. No intervention has a positive registered cardinal lower bound because at least one access or non-overlap factor remains unbounded; this fail-closed accounting floor is not evidence of zero educational benefit. The headline portfolio therefore alternates explicitly labelled base, optimistic, and scarcity sensitivities rather than fabricating a strict conservative order. An earlier build wrongly excluded Bahasa Indonesia because existing work placed it in a completion-only bookkeeping class. On the same ex-ante basis used for every other natural-language candidate, the official 2022 BPS functional-language ceiling places Bahasa Indonesia first in the base and optimistic views. The corrected Top 10 is Bahasa Indonesia, Bangladesh Bangla, Telugu, Indian Bengali, Vietnamese, Marathi, Indian Tamil, Western Punjabi, Javanese, and Gujarati. The first needs-audited products range from an exact-gap audit of the existing Indonesian program and Bangla/Tamil/Gujarati foundational-recovery packages to a Vietnamese precalculus-to-undergraduate spine; Japanese school algebra is a negative control, while advanced Japanese algebra and geometry remain a legitimate targeted lane. Costing the 100

selected language positions against the fixed Formal Reasoning Core comparator yields 91.274 million, 407.305 million, or 1.866 billion gross tokens under the low, base, and high workflows; these are workload comparators, not the needs- optimal first packages. An empirical Indonesian audit confirms the distinction: over a defined 2026-08-20 to 2026-08-29 root-event window, 33 user-visible roots recorded 83.639 billion total tokens, 97.714% of gross input was cached, and a broader 6,726-thread descendant closure recorded 10.253 trillion cumulative total tokens. Those nested boundaries and a separate 88.493-million-token pursuit-goal attribution ledger are reported without addition or price conversion. Four shared-core package models with complete component joins show modeled base-scenario savings of 25%-37.5%, but remain noncardinal because their frozen population-link controls do not authorize additive package unions. This lack of a registered demographic multiplier is not a finding of zero mathematical usefulness: formal density and reader expertise make receptive interlanguage use more plausible in advanced mathematics than in novice exposition, and the paper specifies how to measure that breakpoint. Eleven accessibility interventions are preserved in a separate safeguard backlog because their linguistic and disability/connectivity population ceilings overlap. The result is a grant-oriented commissioning order with exact production identities, machine-readable provenance, open-license and one-download offline eligibility, and uncertainty that remains visible rather than being converted into invented precision.

Keywords: educational access; language of instruction; open educational resources; machine translation; low-resource languages; interlanguages; OpenStax; Open Logic Project; accessibility; cost-effectiveness

1. Introduction

The relevant optimization problem is not how many translated files an AI system can produce. It is how much *additional usable educational access* a finite amount of compute can create. Those quantities diverge whenever a nominal language edition is not readable in the relevant variety or script, duplicates comfortable access through another language, contains content above the learner's current level, fails to reach low-bandwidth or disabled readers, or remains detached from instruction and actual use.

Two widely cited global language-of-education statistics illustrate why the denominator must be defined before it is optimized. Walter and Benson's (2012) Table 14.2 (p. 283) estimated that 3,741,110,588 people belonged to languages used in formal education and 2,300,263,716 to languages not used, yielding 61.9% and 38.1% of their covered global population. The same table reported 97 languages with more than ten million speakers: 52 used in education and 45 not used. Those 45 accounted for 1,120,220,125 people, almost half of the population in the table's "not used" side. The chapter prints

size bands, not a list of the 97 languages. It therefore supports a concentration argument - large neglected languages matter - but cannot identify a modern target list.

The World Bank's later estimate is different. It reports that approximately 37% of students in low- and middle-income countries, about 370 million of roughly one billion students, are taught in a language other than their first language or the language they best speak and understand (World Bank, 2021). This is a student and instructional-language estimate, not Walter and Benson's global language-population calculation. Neither percentage can be multiplied by every named language population to manufacture the number a translation would reach.

The underlying distribution nevertheless makes allocation consequential. In Walter and Benson's (2012) table, the 45 noneducation languages above ten million and the 662 in the 250,000-9,999,999 band together represented 2,088,576,471 people - 90.8% of the population assigned to languages not used in education. A compute program can thus combine two aims rather than choose between them: large exact-language editions can deliver enormous aggregate reach, while smaller endangered, signed, Indigenous, and minoritized-language editions can expand public function, regional depth, and prestige domains that a population-only ranking erases.

AI changes the feasible frontier but not the definition of access. Multilingual models, terminology constraints, deterministic source pipelines, independent model critique, and formula/structure checks make many editions cheaper and more auditable. Yet low-resource model performance, corpus quality, dialect coverage, and specialist terminology remain uneven (Blasi et al., 2022; Dinu et al., 2019; Kantharuban et al., 2023; Kreutzer et al., 2022; NLLB Team, 2024; Robinson et al., 2023). A global portfolio therefore needs exact target identities, an explicit overlap model, and uncertainty that remains visible rather than being compressed into an opaque score.

2. Research questions

The study asks:

1. Which exact language, variety, script, territory, and accessibility interventions have the largest dated source-population ceilings among communities not already represented by the local Open Logic/OpenStax portfolio?
2. How do those ceilings change after directly evidenced or bounded deductions for literacy/modality access, academic-lingua-franca access, and existing editions?
3. Which open curriculum package provides the most defensible first educational intervention for each population, using Open Logic and OpenStax as measured baselines rather than as an exhaustive source universe?

4. How many low/base/high inference tokens and API-equivalent dollars would those packages require under a reproducible multi-pass translation and audit workflow?
5. When can an interlanguage, pluricentric semantic core, dual-script package, or natural-intercomprehension workflow improve reach per compute relative to separate editions?
6. Which exact natural-language or shared-core interventions form a deduplicated Top 10 and Top 100 under reach-first, scarcity, and overlap sensitivities, and which accessibility mechanisms must be retained as separate horizontal safeguards because their residual beneficiary counts cannot yet be ranked cardinally?

3. Conceptual framework

3.1 Access is not learning, but it is a necessary input

Language adaptation can remove a barrier without guaranteeing learning. A translated edition must still be instructionally appropriate, delivered, used, and integrated with practice. This paper therefore ranks **expected marginal educational access** rather than test-score effects. Learning-adjusted benefits require a specified deployment and appear only as a later scenario.

This choice also follows comparative cost-effectiveness practice: a common numerator is useful only when its outcome definition and cost boundary remain explicit (Angrist et al., 2025; Dhaliwal et al., 2013). Here the common numerator is access created, not a synthetic learning-effect estimate.

For population cell c , intervention i , and curriculum unit k :

$$EMA_{cik} = N_c D_{ck} C_{ci} P_{ci} U_{ci} R_{ck}$$

where N is a dated source population; D is the missing-content share; C is comprehension of the proposed linguistic surface; P is practical written, signed, device, bandwidth, and disability access; U is residual non-overlap; and R is relevance of the fixed curriculum to the specified learner stratum. Scarcity, vitality/equity, prestige-domain value, production feasibility, and dialect-flattening risk remain separate views rather than invisible multipliers of people.

This extends the recovered local marginal-intelligibility formalism (*Crosswalk to earlier local marginal-intelligibility research*, n.d.; *Marginal-intelligibility opportunity heuristic for OpenLogic*, n.d.). Its core opportunity dimensions are newly comfortable readers (R), scarcity (S), practical accessibility (A), and non-overlap (N), with vitality/prestige/feasibility/risk kept separate. Numeric intervals refine those bands without redefining them.

3.2 Three quantities that must not be conflated

Every language table distinguishes:

- **source observation:** exactly what the source counted, including year, unit, and measure type;
- **new-edition access ceiling:** the compatible written or modality population for an exact target; and
- **marginal comfortable-access range:** the residual after access through existing editions and academic lingua francas is deducted.

A mother-tongue census count, a household-language count, self-reported speaking ability, an ethnic-group population, a refugee caseload, and school enrolment are not interchangeable. Aggregate Quechua, Nahuatl, Maya, Cree, Ojibwe, Inuktitut, Fula, Kurdish, Amazigh, or Romani labels are not assigned to a single edition surface.

3.3 Interlanguage mechanisms

“Interlanguage” covers mechanisms with different numerators and denominators:

1. A **constructed bridge surface** seeks receptive comprehension across named communities. It earns reach only from evidence for that surface, script, cohort, and task.
2. A **pluricentric shared semantic core** generates explicitly named localized outputs. Reach belongs to each native-language output; the shared source reduces compute.
3. **Natural intercomprehension reuse** shares terminology, source structure, and QA across related languages while preserving separate editions.
4. A **dual-script/register package** produces synchronized outputs for one language continuum without adding script populations twice.
5. An **accessibility derivative** removes a format, device, register, or disability barrier within a language population and never substitutes for linguistic access.

This taxonomy prevents a family name from turning into hundreds of millions of unsupported readers. The current Interslavic evidence includes an approximately .84 pooled result on a seven-gap short written cloze task, but no low/high interval and no demonstration of sustained mathematical reading. The pooled point is retained as an upper-bound research sensitivity; it is not inherited by Russian, Ukrainian, Polish, Czech, Slovak, South Slavic, or minority-language leaves (Merunka et al., 2019).

3.4 Mathematics changes the interlanguage prior

The absence of a registered population multiplier is not evidence that an interlanguage is useless. Mathematics is a mixed semiotic system: prose, symbols, diagrams, layout, and previously learned formal structures jointly carry meaning. Expert mathematicians reading unfamiliar advanced material use disciplinary knowledge, examples, self-checks, and structural expectations differently from less

experienced readers (Shepherd & van de Sande, 2014). Neurocognitive evidence also separates high-level mathematical reasoning from ordinary-language processing in important respects (Amalric & Dehaene, 2016). These findings make cross-language or interlanguage reading more plausible in mathematically expert, symbol-dense settings than it would be for ordinary expository prose.

The effect cannot be generalized downward. Mathematical textbooks continually ask readers to switch between words and symbols; in a 180-page stratified textbook sample, 4,242 symbolic structures were coded in 180 randomly selected textbook pages, and 82% were connected to natural language (Wikström Hultdin et al., 2023). Symbolic mathematical text can itself be difficult for students to comprehend (Österholm, 2006), and word problems, definitions, motivation, and error explanations remain linguistically dense. Accordingly, the model uses an **expertise x formal-density prior**:

- foundational and remedial material receives no assumed cross-language benefit;
- secondary and undergraduate material is task-dependent;
- advanced/reference material receives a higher research prior for receptive interlanguage usefulness, but only measured task evidence can turn that prior into a cardinal beneficiary count.

This breakpoint makes interlanguages a serious mathematical research and production hypothesis without turning a family name into automatic comprehension.

4. Evidence from education, OER, AI, and accessibility

4.1 Familiar-language instruction can help, with heterogeneous effects

Nakamura et al.'s (2023) systematic review included 45 studies and found positive pooled effects for several literacy outcomes: letter knowledge, sentence reading, reading comprehension, and national-language reading comprehension. Word reading was near zero, and writing evidence rested on one study. The result supports language as an enabling condition, not one transportable effect size.

Setting-specific evidence shows both upside and implementation risk. A Kom-medium program in 24 Cameroonian schools reported large early gains and persistent later gains after transition to English (Laitin et al., 2019). South African administrative panel evidence associated early transition to English with lower later English achievement than three years of home-language instruction (Taylor & von Fintel, 2016). Conversely, a Kenyan trial adding Lubukusu or Kikamba literacy to an English/Kiswahili package did not improve the wider-language outcomes and produced lower mathematics results in that implementation; limited classroom use, teacher language mismatch, and resistance mattered (Piper, Zuilkowski, Kwayumba, et al., 2018).

One official language-specific administrative example shows why the global 37%-40% figures cannot simply be assigned to every language. South Africa's 2016 school tabulation reported 2,402,540 Foundation Phase learners whose school offered a language of learning and teaching matching home language and 540,484 whose school did not; in Intermediate Phase the corresponding counts were 585,808 and 1,844,086 (Department of Basic Education, 2023). These are phase-specific principal-reported EMIS cells with verification warnings, not a global causal estimate or a count that can be multiplied by the 2022 census. They do, however, identify a concrete cohort and show how medium-of-instruction mismatch can rise sharply after the early grades.

The allocation consequence is precise: familiar-language evidence justifies a linguistic-usability term and exact cohort research. It does not justify multiplying a static PDF by a generic standardized-effect coefficient.

4.2 Teacher-independent packages and classroom packages are complementary

The target user is not assumed to have a teacher. A motivated learner who can obtain the package once should be able to place themselves, follow a prerequisite path, study worked examples, attempt exercises, diagnose errors, and check complete public solutions offline. Worked examples can reduce unnecessary search while learners acquire algebraic schemas (Sweller & Cooper, 1985). Direct learner-facing digital mathematics interventions have also produced gains in settings with limited resources, including a randomized tablet intervention in Malawi (Pitchford, 2015). Neither result proves that a book alone teaches everyone; it supports treating diagnostics, examples, practice, and feedback as part of the product rather than assuming a teacher supplies them.

Classroom adoption remains a complementary route. In a Kenyan randomized evaluation, professional development, coaching, student books, and structured teacher guides operated as a package and produced large gains; teacher guides added little cost relative to the other components in that setting (Piper, Zuilkowski, Dubeck, et al., 2018). McEwan's (2015) meta-analysis found positive average effects across several intervention classes but also showed that composite treatments and sparse cost data complicate category rankings.

Input-only failures matter. Textbooks in 100 Kenyan schools had no average effect and benefited mainly higher-performing pupils; the books were in English and pitched to a demanding curriculum (Glewwe et al., 2009). In Sierra Leone, many supplied books remained stored and no learning effect was detected (Sabarwal et al., 2014). The Global Education Evidence Advisory Panel (2023) accordingly treats structured pedagogy and teaching at the right level more favorably than unintegrated inputs.

Each recommended edition therefore names a bounded curriculum and derivative set: editable source, preserved examples/exercises, diagnostic placement, hints, public answers, worked solutions and

misconception explanations, accessible HTML/MathML, visually checked screen and print PDFs, and one-download offline distribution. A teacher guide can be added for classroom reuse; it is not the missing engine that makes the learner edition coherent.

4.3 Openness is an eligibility condition for a global translation program

Open textbook meta-analysis finds little average achievement difference between open and commercial textbooks, with some evidence of lower withdrawal and strong access and cost advantages (Clinton & Khan, 2019). Reviews and observational studies are generally compatible with “no worse learning, lower access barriers,” but most are not randomized and are concentrated in North American higher education (Colvard et al., 2018; Hilton, 2016).

Grimaldi et al. (2019) formalize the access hypothesis: when only students previously unable to obtain a commercial text benefit, whole-class averages dilute the effect. That mechanism motivates the present numerator - newly usable access rather than all speakers, downloads, or enrolments. For the present program, lawful openness is not a minor predictive factor: it is an eligibility condition. A global many-language workflow must be able to translate, adapt, correct, redistribute, print, preserve, and revise the source without charging the learner or renegotiating each edition. UNESCO's (2019) OER Recommendation explicitly supports open licensing, adaptation, translation, redistribution, accessible formats, and offline/printed access.

This is compatible with the empirical finding that open and commercial textbooks often have similar average classroom outcomes. The claim is operational, not magical: openness removes price and permission barriers, permits one source improvement to propagate across languages, and prevents schools or independent learners from being locked into a purchase. “Free to view” is recorded as existing supply but is not treated as reusable OER unless the exact artifact license permits the required work.

4.4 The translating model must also understand the mathematics

NLLB demonstrates the engineering value of multilingual transfer, but aggregate benchmark gains coexist with large resource-level disparities (NLLB Team, 2024; Robinson et al., 2023). Corpus audits show why nominal data volume can mislead: Kreutzer et al. (2022) found that many web-crawled language corpora contained less than half usable text and some contained no in-language material in the audited sample. Dialect performance gaps are substantial and inconsistent (Kantharuban et al., 2023); relatively small dialect-specific parallel corpora have produced large improvements over standard-language-oriented systems in earlier MT settings (Zbib et al., 2012).

The relevant intervention is not a cheap general machine-translation call. It is a reasoning model capable of reconstructing the mathematics, detecting whether a definition, quantifier, implication, index, or formula has changed, and revising the target prose in light of that semantic check. Scientific-paper translation studies show that LLM workflows can preserve a high proportion of tested source details while still producing terminology and overtranslation errors (Kleidermacher & Zou, 2025). MathMist likewise finds substantial cross-language degradation in mathematical reasoning, especially for lower-resource languages (E Sobhani et al., 2026). Capability therefore changes the workflow, not the need for evidence.

Terminology constraints raise exact term-use rates but do not guarantee grammar, semantics, or mathematical fidelity (Bergmanis & Pinnis, 2021; Dinu et al., 2019). The compute model budgets a mathematically capable translation pass, an independent model critique/reconstruction pass, terminology reconciliation, literal source and formula comparison, structural QA, build/render inspection, correction cycles, accessibility derivation, and retry recovery. Symbolic-equivalence and semantic-consistency selection can improve mathematical autoformalization (Li et al., 2024), so formalizable fragments receive additional deterministic checks. Human feedback can inform later revisions, but no human-dependent approval gate is required for a reversible, source-audited release.

4.5 Accessibility and low bandwidth are separate marginal-access axes

WCAG 2.2, EPUB Accessibility 1.1, and MathML Core provide machine-testable and judgment-dependent requirements for web, publication, and mathematical structure (W3C Math Working Group, 2025; World Wide Web Consortium, 2024a, 2024b). Conformance is not the same as successful use across every assistive-technology stack, but native semantic source avoids repeatedly remediating inaccessible PDFs.

Low-tech evidence again emphasizes intervention design. Phone tutoring plus SMS improved numeracy in Botswana, while SMS alone did not (Angrist et al., 2022). A Sierra Leone phone-tutoring intervention did not improve language or mathematics amid limited fidelity and take-up (Crawford et al., 2023). Offline HTML/EPUB, downloadable packages, audio, plain language, Braille, and signed-language video are therefore costed as specific derivatives, not credited merely because a file can be sent over a low-bandwidth channel.

The baseline distribution hypothesis is intermittent rather than continuous access: someone with a phone and one successful connection should be able to acquire the course. The package therefore avoids login, streaming, and live-link dependencies; supports shared devices, low storage and resumable chapter downloads; and carries its answers, fonts, glossary, prerequisite map and version information locally. SMS/USSD can support placement, reminders, and short drills, but cannot carry the diagrams, fractions, derivations, and proofs that define a complete mathematics course.

5. Data

5.1 Population observations

The population master preserves source labels, years, measure definitions, units, territories, exact counts or source intervals, nesting, and alternative-measure groups. Official censuses and statistical reports are preferred. Where an official source lacks direct language measurement, an ethnic group is retained only as a proxy/context row and cannot enter the language ranking. Survey case frequencies are not national populations. Same-name Indian C-16 mother-tongue components are used in preference to broader scheduled-language categories; C-17 English/Hindi knowledge remains a category-level overlap sensitivity.

Where a direct national count was not registered, selected large-language ceilings use the pinned CLDR 48.2 population XML and transparent derivation chain. CLDR's heterogeneous estimates have unspecified component vintages and intervals, so these rows remain secondary ceilings rather than census observations (Unicode Consortium CLDR Technical Committee, n.d., 2025, 2026b).

The final frozen inventory and region/source counts are inserted by the build:

Table 1. Frozen analytical inventory and cardinal treatment.

Registered layer	Rows	Cardinal treatment
Dated population observations	475	Source facts; heterogeneous measures are never summed globally
Registered population-source records	81	58 distinct authority labels; provenance and limitation register
Candidate intervention hypotheses	210	Discovery universe, including unrankable hypotheses
Scored exact natural-language/profile rows	143	Prior rows plus globally balanced expansion and the equal-basis Bahasa Indonesia correction
Eligible exact natural-language interventions	134	Headline cardinal inventory
Richly served negative controls	4	Excluded from generic translation priority
Unresolved or D0 profile exclusions	5	Three unresolved profiles and two population/profile mismatches
Intervention-matrix rows	104	80 interlanguage plus 24 accessibility rows; not automatically cardinal
Interlanguage population links	113	All remain nonadditive under the frozen upstream gate
Package × token × reuse sensitivity rows	135	Engineering comparison only; zero packages enter the headline ranking
Accessibility safeguard interventions	11	Separate noncardinal backlog; zero numbered linguistic positions

Note. Row counts describe registered records, not unique people. The 104 intervention-matrix rows include both interlanguage and accessibility mechanisms; the 80-row interlanguage subset is summarized separately in Appendix H.

5.2 Existing local editions and work in progress

The bounded successor-state census verifies complete Open Logic baselines for zh-Hans-CN, hi-Deva-IN, tr-TR, Modern Standard Arabic, fa-Arab-IR, Spanish, pt-BR, and Bahasa Indonesia; the Chinese lane also has all 94 *Algebra and Trigonometry 2e* modules and three of 55 *Calculus Volume 1* modules at its pinned state. The public Indonesian Open Logic repository independently identifies the frozen closure as 722/722 editable content units and its linked reader as 1,116 pages. The canonical reader reaches 642 modules; the other 80 translated targets are non-reader source modules retained in the editable package rather than missing translations (OpenLogic-id, 2026). It therefore distinguishes complete Indonesian coverage from partial Interslavic work and from provisional or nonadmitted bridge artifacts. A complete exact edition is baseline coverage, not a new intervention: for any forward Open Logic allocation to Indonesian, the content deficit is ($D=0$), and the already delivered units must not be counted as future benefit or future translation workload.

Two estimands are now kept separate. The **ex-ante equal-treatment language opportunity** asks whether an exact Bahasa Indonesia edition belonged in the original candidate set before any local work existed; for that comparison ($D=1$), exactly as for every other then-missing FR-2 edition. The **forward allocation** asks what useful content or access layer remains after the existing Indonesian program. That second question starts from ($D=0$) for Open Logic and applies unit-level deficits only to specific, demonstrably absent curriculum or accessibility components. Treating prior work as a reason to exclude Bahasa Indonesia from the first comparison was a classification error; treating the completed 722-unit edition as unfinished would be the opposite error.

The Indonesian work is an established mathematics program, not a pilot. At the pinned public program commit audited for this revision, all 40 course roles have a selected corpus or frozen original specification. Applying the repository's executable live publication overlay to its base course map yields 27 published roles and 13 production roles: 67.5% complete by course-role count, with no planned or unresolved roles (Program Matematika Indonesia, 2026). Page counts require a stricter distinction. The central evidence ledger records 19,745 measured teaching-package pages, 20,763 selected-corpus working pages, and a 27,705-page documented rendered universe that includes 306 donor pages and 7,654 Stacks reference pages. A separate pinned-public- artifact reconstruction yields 26,031 de-duplicated rendered course/checkpoint pages. These figures establish the program's scale, but none is a final translated-page total; “about 25,000 pages” is therefore retained only as an order-of-magnitude description whose exact page universe must be named. The public catalog also records a complete 82/82-module *Elementary Algebra 2e* edition, so neither another Prealgebra translation nor the formerly recommended Elementary-Algebra bridge is a valid next commission. The correct forward rule is to inventory the live program against the needs layer, finish only the exact missing or in-production roles, and improve placement, solution coverage, offline delivery, and accessibility where the existing

edition evidence shows a real gap. This does not change the ex-ante result: Bahasa Indonesia still ranks first in the equal-treatment language opportunity set.

5.3 Curriculum denominators

The principal comparator is the complete **FR-2 Formal Reasoning Core** at D3: 210 editable units and 120,083 measured source alpha tokens. Independent remeasurement verified every one of 722 Open Logic source units and the complete 367,220-token closure. The pinned source identity is OpenLogic commit `1e960beff9ed7835bf3e3f1335e21af3439cd107`, licensed CC BY 4.0 unless a component states otherwise (Open Logic Project, 2026).

OpenStax supplies concrete expansion portfolios rather than a vague content category. The locally available measured set comprises 257 modules and 1,017,337 source tokens: Algebra and Trigonometry 2e (94 modules; 394,473), Calculus Volume 1 (55; 195,458), Calculus Volume 2 (54; 180,155), and Calculus Volume 3 (54; 247,251). Four further books have exact module counts but only planning token intervals and are never misrepresented as measured. Fixed-source compute comparators choose among a minimum viable quantitative gateway, statistics/business, science bridge, undergraduate STEM, and formal-reasoning portfolios, plus accessibility derivatives. The needs recommendations are selected separately from learner-stage evidence; Appendix F's common-book mapping is not relabelled as a local-needs finding. Appendices A and F identify the exact collection commits, licenses, row-level curriculum measurements, and compute artifacts behind every denominator (OpenStax, n.d.-a, n.d.-b, n.d.-c, n.d.-d, n.d.-e, n.d.-f, n.d.-g, n.d.-h).

5.4 Scarcity, digital context, and model support

OpenAlex counts of mathematical works and open-access books are retained as a bibliographic scarcity proxy (OpenAlex, 2026). Wikimedia project counts provide broad digital-presence context (Numberof bot, 2026). Country internet, electricity, literacy, and enrolment indicators provide delivery sensitivities (World Bank, 2026). FLORES/SONAR and NLLB resource levels provide model-support context (Duquenne et al., 2023; NLLB Team, 2024). None of these is relabelled as an inventory of usable advanced teaching material, a language-specific accessibility rate, or proof of translation quality.

5.5 Population-specific mathematics-needs register

The content-needs unit is language x territory x learner stage x delivery context. The audited register combines primary or official mathematics outcomes with known curriculum/OER supply and delivery constraints for the corrected Top 10 plus a Japanese calibration case. It distinguishes four first-product tiers:

1. foundational recovery: diagnostic number sense, operations, fractions, proportional reasoning and prealgebra with complete feedback;
2. secondary bridge: algebra, functions, trigonometry, data and transition into a later academic language where relevant;
3. undergraduate spine: coherent precalculus, calculus, linear algebra, discrete mathematics and proof rather than disconnected modules; and
4. advanced/reference: a documented dependency ladder for a specialist cohort.

The first three tiers can coexist within one language; the model selects the lowest binding prerequisite for each observed cohort rather than branding a whole population “foundational.” ASER state figures are rural territorial proxies, PISA covers enrolled 15-year-olds with explicit coverage rates, and Indonesia's national outcomes are not Javanese-specific. Those scope limits are preserved row by row in `population_mathematics_needs_register.csv`.

Japan supplies a useful negative control on generic recommendations. PISA 2022 reports that 88% of represented Japanese 15-year-olds reached mathematics Level 2 or higher and 23% reached Levels 5-6 (Organisation for Economic Co-operation and Development, 2023c); the national government supplies compulsory-school textbooks free of charge (Ministry of Education, Culture, Sports, Science and Technology, n.d.). Duplicating mainstream school algebra therefore has low marginal priority. That does not make Japanese mathematical translation useless: a proof-to-abstract-algebra-to-commutative-algebra-to-algebraic-geometry sequence addresses a different, narrower advanced and reference gap. The distinction is content-specific, not a verdict on the language.

6. Methods

6.1 Exact intervention edge construction

Machine joins use exact ISO/code crosswalks, normalized explicit aliases, target territory, and source measure. Code-only matches between distinct profiles - for example Tetun Prasa and Tetun Terik - are routed to review and never accepted automatically. Parent/child alternatives, bilingual categories, script variants, household measures, oral ability, and proficiency measures retain nonadditive overlap groups. Every final Top 10 and Top 100 entry requires a rankable exact production target and a registered person observation copied from a dated authority or calculated by a fully exposed derivation from dated inputs. The population measure need not equal the production audience: where a census label pools standards, communities, or orthographies, the count is retained once as a ceiling and the mismatch is stated. Unresolved macro-targets appear in a separate research appendix.

6.2 Factor assignment

For a missing exact natural-language FR-2 edition, $D=1$. For an exact named surface, $C=1$ is a target-definition identity conditional on literacy in that standard - not a claim that every speaker understands mathematics. Language-specific literacy is used where reported; otherwise a national literacy value appears only in a wide proxy sensitivity with low 0 and high 1. Internet and electricity select delivery modes rather than comprehension.

The equal-treatment ranking freezes the decision date before candidate-specific production and therefore applies the same ($D=1$) comparator to Bahasa Indonesia and every other language. A separate forward-looking table starts from current coverage and substitutes the exact remaining-content share and incremental compute. This counterfactual/current split prevents successful programs from being penalized merely because work already exists, while still preventing completed bytes from being counted twice as future benefit.

Non-overlap is factored into existing-edition, academic-lingua-franca, and selected- portfolio residuals. India C-17 supports a disclosed sensitivity (Office of the Registrar General & Census Commissioner, India, 2018): the low scenario treats every reported English-or-Hindi speaker as already served, the base treats half as comfortably served, and the high treats one fifth as served. Elsewhere, unknown academic overlap is $[0, 1]$; a value of .5 appears only in a neutral sensitivity, not as observed evidence.

The implemented paper consequently reports five views: exact gross ceiling per common compute; the direct-evidence marginal floor; base and optimistic factor-model sensitivities; a scarcity-adjusted sensitivity; and fail-closed interlanguage architecture comparisons. Vitality/equity, prestige, feasibility, and dialect risk remain descriptive or eligibility evidence; the frozen selector does not manufacture numeric portfolio objectives for them.

6.3 Compute scenarios

The workload ledger separately counts uncached input, cached input, and output. It does not recover or model cache-write tokens as a separate category; the API-equivalent dollar scenarios therefore exclude any cache-write surcharge. Gross, fresh, cached, weekly-plan, and monetary quantities are not interchangeable. For FR-2 D3, the low/base/high scenarios are:

Compute comparator. One exact FR-2/D3 edition under the three planning workflows.

Scenario	Gross tokens	API-equivalent USD	Status
Low	912,737	5.19	Planning workflow
Base	4,073,049	16.77	Planning workflow
High	18,664,571	65.43	Stress workflow

These dated dollar values are API-equivalent comparisons under the recorded GPT-5.6 Sol prices - \$4 per million uncached input tokens, \$0.40 per million cached input tokens, and \$20 per million output tokens - accessed August 25, 2026 (OpenAI, 2026). They are not measured weekly Codex-plan consumption. The base scenario gives every unit an independent critique and allows substantial correction; the high scenario adds multiple broad passes and retry allowance. The full 722-unit Open Logic corpus costs 2.679/11.666/53.200 million gross tokens in the same low/base/high model.

The earlier Indonesian “remainder” estimate is retired: the public Indonesian Open Logic closure is complete at 722/722, so its forward translation workload is zero. Program-scale compute must instead be measured against specific unfinished course or accessibility components. The empirical task-accounting subsection below distinguishes pursuit-accounting tokens, gross input, cached input, uncached input, output, reasoning, requests, and monetary cost. It reports only fields actually preserved by the tasks and does not invent a component split or price conversion.

6.3.1 Empirical Indonesian production accounting

A central audit of the manager plus 32 user-visible owner roots covers root events from 2026-08-20T10:40:19Z through 2026-08-29T23:33:38.559Z. All 33 final rollout totals match the corresponding `state_5.sqlite` counters. At this boundary, gross input is 83,386,749,267 tokens, of which 81,480,422,656 (97.714%) is cached and 1,906,326,611 (2.286%) is fresh/uncached; output is 251,883,504, including 70,812,538 reasoning tokens that are already a subset of output. Total usage is 83,638,632,771 tokens. Cache-write input is reported as zero. Reasoning is not added a second time.

Thirty-three-root field	Exact tokens	Accounting rule
Gross input	83,386,749,267	Cached plus fresh input
Cached input	81,480,422,656	97.714% of gross input
Fresh/uncached input	1,906,326,611	2.286% of gross input
Cache-write input	0	Exact reported field at this boundary
Output	251,883,504	Includes reasoning output
Reasoning output	70,812,538	Subset of output; never added again
Total	83,638,632,771	Gross input plus output

The descendant-inclusive closure is much larger: 6,726 thread records and an exact cumulative total-token counter of **10,253,232,856,362** through 2026-08-29T23:34:32Z. Its historical cached/fresh/cache-write/output/reasoning split and full request count are unavailable. A bounded final-rollout scan finds 7,970 distinct cumulative-token progressions, which is only a request/progression lower bound. The sanitized audit receipt proves that the recursive closure begins with and therefore already contains the 33 roots. It is an inclusive accounting boundary: it may be compared with the 33-root subtotal but is never added to it, and no descendant- exclusive remainder is asserted.

This measurement is not the whole historical Indonesian program. Open Logic work before 2026-08-20 is excluded because registry 152 does not map it to a canonical task ID; auxiliary audit and research tasks outside the selected 33 roots are excluded; and requests without surviving counters are excluded. These omissions are an unknown positive remainder, not zero.

A separate bounded audit of surviving pursuit-goal receipts attributes **88,493,496 Codex pursuit-accounting tokens** to nine nonoverlapping phases. Of that amount, 87,035,193 belongs to edition production, translation, mathematical/terminological QA, build, and publication goals; 1,458,303 belongs to separately identified deployment, maintenance, and durable-state correction goals. This is a strict measured lower bound: it is a narrow workflow-attribution ledger inside the larger program activity, not a program total. Central coordination, most other course tasks, work outside an active pursuit goal, and several known helper/revision phases have no surviving comparable goal counter.

Measured task phase	Exact output boundary	Pursuit-accounting tokens	Elapsed counter	Classification
<i>Methods in Algebra, Volume 2</i> complete edition	146 units; 864 pages; public commit 8dbaeb4443978aef6d89365149e28a6ba06e005a	31,014,565	approximately 49 h 21 min	Edition production
Same edition, later Pages deployment	31 reader files; deployment-only commit 7aacf53215171cfc734e963bdc40ac8f3ed dfe13	586,587	30 min 28 s	Support; no corpus change
<i>Elementary Algebra 2e</i> owner checkpoint	82/82 modules; current 2,158-page release boundary	38,663,209	69 h 40 min	Live lower bound for edition production
<i>Elementary Algebra 2e</i> 17-module helper	17-module packet; 869 files independently replayed	2,022,997	approximately 3 h 13 min	Edition-production helper
<i>Applied Combinatorics</i> production	Public commit 50cb1c9eae0273d7235494c747555be2b4e9f910	10,838,830	16 h 39 min 8 s	Edition production
<i>Applied Combinatorics</i> maintenance/publication	Same public lineage; separate later goal	832,315	1 h 23 min 40 s	Support
CLP4 Indonesian helper	316-page text plus 486-page problem book; 4,477 files	2,683,906	4 h 21 min 21 s	Edition-production helper

Measured task phase	Exact output boundary	Pursuit-accounting tokens	Elapsed counter	Classification
<i>Elementary Algebra 2e</i> 15-module replacement helper	15-module packet; 843 retained files checksum-closed	1,811,686	54 min 58 s	Edition-production helper
<i>Elementary Algebra 2e</i> durable-goal correction	Cursor/handoff reseal; no translation change	39,401	4 min 6 s	Support
Measured lower bound	Nine nonoverlapping task-goal phases	88,493,496	Not summed: work ran in parallel	Known remainder is positive

The Codex goal implementation defines the pursuit counter as input tokens minus cached-input tokens plus output tokens, and descendant-agent usage rolls into the root pursuit (OpenAI Codex, 2026). It is therefore neither gross input nor a separate sum of parent and subagent counters. The pursuit receipts do not preserve the component totals needed to reconstruct gross input or API-equivalent cost; the 33-root audit supplies a split for a different, wider boundary, while the 6,726-thread closure supplies only total tokens. None of the three views is an invoice, a weekly-plan percentage, or a token-per-page conversion rule, and they are never added. Token totals are workload and accounting proxies, not FLOPs, hardware time, or energy. Together they demonstrate that the paper's source-length planning scenarios are far below observed end-to-end program workflows once mathematical checking, terminology, deterministic replay, accessible reader/PDF/backend construction, repair, and publication are included.

6.3.2 Role-count rescaling and next-language sensitivity (not a forecast)

The current public role boundary is 27 published of 40, but the observed counters already include activity from both published and production roles, and the 13 production roles have consumed unknown compute and differ sharply in size. There is therefore no matched completed-role denominator from which to estimate remaining compute. Multiplying the counters by 40/27 is retained only as a deliberately crude arithmetic replay-scale screen: it gives 123.909 billion tokens at the 33-root total boundary, 3.197 billion at the derived root fresh-input-plus-output boundary, and 15.190 trillion at the descendant-closure total boundary. These are not completion estimates, remaining-work estimates, or second-language forecasts; they simply expose the consequence of an explicitly arbitrary role-count assumption.

For a second language using the same selected corpora and build infrastructure, the share of reusable non-language-specific work is unknown. Table 4 therefore applies a reuse sensitivity to the deliberately crude 40/27 replay scale. It assumes no language-difficulty adjustment and is not a price, promise, completion estimate, or forecast.

Table 4. Pure arithmetic replay-scale sensitivity for a subsequent language.

Shared-pipeline reuse sensitivity	Descendant-total-token scale	Root fresh-input-plus-output scale	Interpretation
0%	15,189,974,602,018	3,197,348,319	Full replay sensitivity; no reuse credit
25%	11,392,480,951,513	2,398,011,239	Limited source/build reuse
50%	7,594,987,301,009	1,598,674,159	Half of measured workflow treated as reusable
75%	3,797,493,650,504	799,337,080	Aggressive reuse stress case

The correct next-language forecast must replace this table with task-stage instrumentation: source preparation, semantic translation, mathematical/terminological QA, repair, accessibility, build, and publication counters kept separately. Language-specific difficulty can increase the residual even when source and backend work are reused. Until those stage counters exist, the defensible claim is an observed scale and a sensitivity band, not a single “tokens per page” coefficient.

6.4 Portfolio selection and double counting

The selector operates on canonical intervention cells rather than adding every available language count. Upstream overlap groups choose one observation when a parent category and same-name component are alternative measures. A shared-core package can replace enumerated natural-edition rows only if the validated population-link and intervention-matrix layers explicitly authorize their union. The current

frozen interlanguage model authorizes none: all raw package links remain nonadditive and all candidate matrix rows remain nonrankable. The headline portfolio therefore retains the natural-edition rows and reports package aggregation separately as an engineering sensitivity. Multilingual speaker totals, multiple-response language counts, pooled script or orthography audiences, and accessibility strata retain explicit nonadditivity rules. No unobserved cross-language audience is subtracted or added dynamically.

The conservative efficiency view is reported even when every candidate has a zero floor. Because that view is then a complete tie and supplies no ordering information, admission cycles deterministically through three informative views: base marginal access per million base tokens, optimistic access per million low-scenario tokens, and scarcity-adjusted base access per million base tokens. Ties receive rank intervals. The cycle repeats until every eligible intervention is exposed; the first 10 and first 100 positions form the headline portfolios. Accessibility mechanisms remain in a separate safeguard backlog until a residual population gain can be bounded, rather than filling linguistic positions with global or nested ceilings.

6.5 Interlanguage compute comparison

A constructed surface receives no demographic reach merely from a family name. It can enter a future cardinal comparison only after an exact target/task comprehension gain is registered; under the present conservative rule, constructed Interslavic therefore has a **zero registered cardinal lower bound**. That is an accounting floor, not an estimate of zero mathematical usefulness or zero receptive readability. A shared semantic core is different: it can produce a fully named native-language or script-localized output for every constituent, but shared-source reuse changes compute rather than readership. The analysis therefore compares complete named-output bundles with their independent-production counterparts while preserving each constituent population cell separately. Arithmetic reconciliation of those cells is necessary but not sufficient for a cardinal union; the frozen upstream nonadditivity controls keep every bundle outside the headline Top 10 and Top 100. The paper varies the adaptation fraction widely because no complete category-resolved local usage ledger establishes one universal reuse rate.

7. Results

7.1 Which populations are most affected?

Table 2 reports all 134 canonical natural-language interventions as source-bounded gross ceilings. These values retain each source's territory, year, age universe, question, and measure; they are not interchangeable estimates of people harmed by language mismatch. The largest base ceilings in the frozen candidate set are:

Table 2. Ten largest source-bounded gross ceilings in the eligible candidate set.

Display order	Intervention	Profile(s)	Gross base ceiling	Measure	Year(s)	Source ID(s)
1	Bahasa Indonesia	id-Latn-ID	248,501,794	official_long_form_census_able_to_speak_indonesian_persons_age5plus	2022	PM-S021
2	Bangladesh Bangla	bn-Beng-BD	165,323,060	derived_cldr_territory_functional_language_users	2026-03-17 CLDR release; component estimate year not stated	ASSEC-S006
3	Indian Bengali	bn-Beng-IN	96,177,835	census_same_name_mother_tongue_component_persons	2011	PM-S001
4	Vietnamese	vi-Latn-VN	89,000,000	peer_reviewed_published_lower_bound_speakers	2016 publication; population estimate vintage not stated	ASSEC-S007
5	Western Punjabi	pnb-Arab-PK	88,915,544	census_mother_tongue_persons	2023	PM-S003
6	Marathi	mr-Deva-IN	82,801,140	census_same_name_mother_tongue_component_persons	2011	PM-S001
7	Telugu	te-Telu-IN	80,912,459	census_same_name_mother_tongue_component_persons	2011	PM-S001
8	Indian Tamil	ta-Taml-IN	68,888,839	census_same_name_mother_tongue_component_persons	2011	PM-S001
9	Javanese	jv-Latn-ID	68,044,660	daily_language_at_home_age5plus_persons	2010	PM-S005
10	Thai	th-Thai-TH	64,080,191	derived_union_of_mutually_exclusive_census_strata_persons	2010-09-01	TH-S001

The gross ceilings cannot answer how many readers newly gain comfortable access. The machine-readable `table3_marginal_access_sensitivity_ranges.csv` therefore keeps low, base, and high

marginal-access values separate. The base and optimistic columns are factor-model sensitivities, not observed harmed-population counts. All 134 low values are zero because at least one access or non-overlap factor lacks a positive direct lower bound. The conservative efficiency view is consequently degenerate: every intervention ties on the interval $[1, 134]$, and that view is reported but not used to fill portfolio positions.

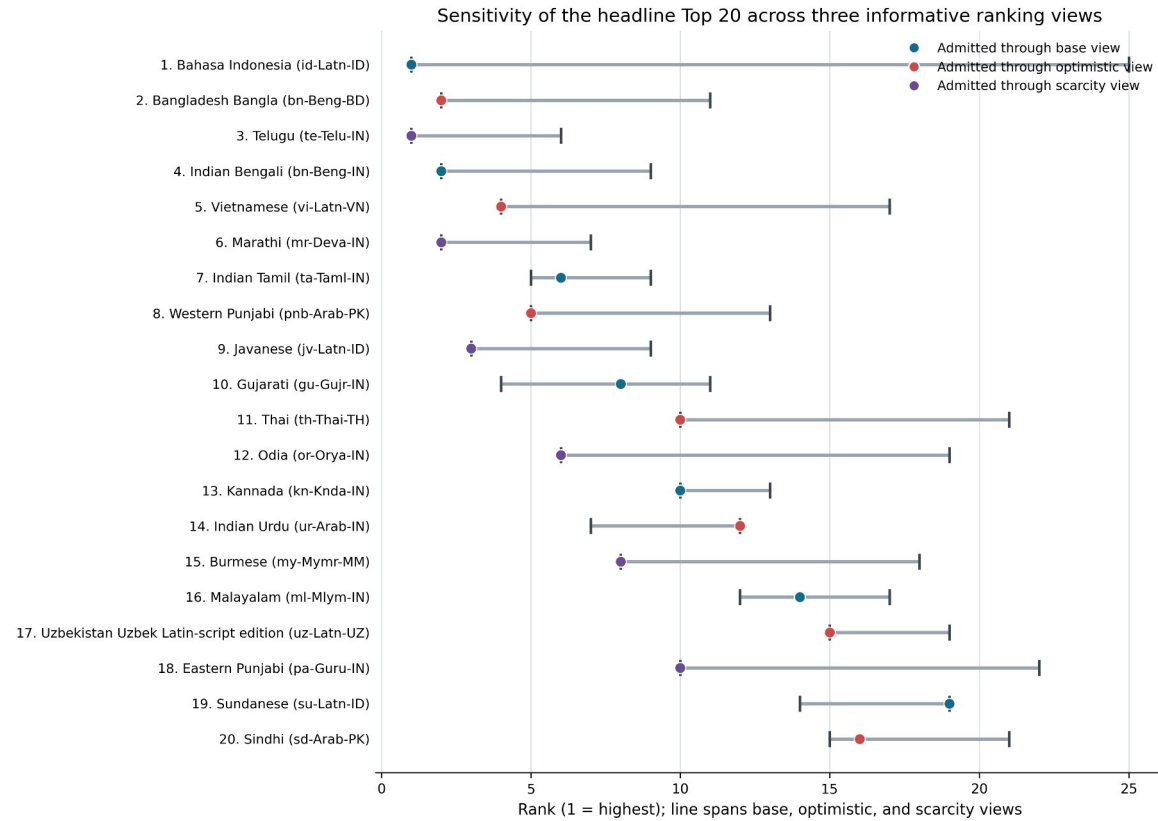
7.2 Top 10 interventions

The selector alternates base, optimistic, and scarcity sensitivity lanes. Under the fail-closed rule, no shared-core package substitutes for its constituent natural-language rows. This is a portfolio order, not a claim of a uniquely identified welfare ranking.

Table 5a. Corrected Top 10 natural-language intervention portfolio.

Position	Intervention	Profile(s)	Gross base ceiling	Base marginal sensitivity	Base gross tokens	Informative rank range	Demonstrated-needs first package
1	Bahasa Indonesia	id-Latn-ID	248,501,794	119,280,861	4,073,049	1-25	Close only verified gaps in the existing 40-course program; do not duplicate complete Open Logic, Prealgebra, or Elementary Algebra editions
2	Bangladesh Bangla	bn-Beng-BD	165,323,060	65,302,609	4,073,049	2-11	Bangla Grade 2-5 foundational numeracy kit
3	Telugu	te-Telu-IN	80,912,459	57,833,869	4,073,049	1-6	Telugu-English Grades 3-10 mastery bridge
4	Indian Bengali	bn-Beng-IN	96,177,835	69,699,712	4,073,049	2-9	Indian Bangla Grades 3-8 arithmetic-to-algebra catch-up
5	Vietnamese	vi-Latn-VN	89,000,000	42,777,849	4,073,049	4-17	Precalculus-to-calculus-to-linear-algebra self-study spine
6	Marathi	mr-Deva-IN	82,801,140	50,316,309	4,073,049	2-7	Marathi Grade 8-to-first-year-STEM bridge
7	Indian Tamil	ta-TamI-IN	68,888,839	48,387,709	4,073,049	5-9	Tamil Grades 2-8 teacher-independent numeracy recovery
8	Western Punjabi	pnb-Arab-PK	88,915,544	26,167,845	4,073,049	5-13	Shahmukhi Punjabi-to-Urdu/English secondary-to-UG bridge

Position	Intervention	Profile(s)	Gross base ceiling	Base marginal sensitivity	Base gross tokens	Informative rank range	Demonstrated-needs first package
9	Javanese	jv-Latn-ID	68,044,660	32,661,437	4,073,049	3-9	Bilingual Javanese-to-Bahasa foundational/academic-register scaffold
10	Gujarati	gu-Gujr-IN	55,036,204	34,089,878	4,073,049	4-11	Gujarati Grades 2-6 diagnostic numeracy remediation



All 134 conservative lower-bound scores are zero and tie at ranks 1-134; that degenerate view is reported separately and is not plotted.

Figure 1. Informative-rank sensitivity for the headline Top 20. The point marks the view that admitted each intervention to the portfolio; the horizontal interval spans its best and worst rank across the three informative views. The conservative lower-bound view is omitted because all 134 interventions tie at zero.

7.3 What mathematics does the Top 10 actually need?

Population rank does not select content. The audited needs layer distinguishes foundational recovery, bilingual transfer, secondary-to-tertiary bridging, a coherent undergraduate spine, and

advanced/reference work. OpenStax and Open Logic remain concrete reusable source baselines, but no project title is assigned merely because its source-token denominator is convenient.

Table 5b. Directly audited Top-10 mathematics needs and first useful open packages.

Intervention	Measured need or best available proxy	First useful open package	Evidence / package confidence
Bahasa Indonesia	PISA: 18% of represented 15-year-old students reached mathematics Level 2 or higher; the live public program already spans 40 selected course roles and includes complete Open Logic and Elementary Algebra baselines	Audit the live prerequisite map; finish exact incomplete foundation/bridge roles and missing placement, solution, offline, or accessibility layers without duplicating completed books	high / medium
Bangladesh Bangla	NSA 2022: 39% of Grade 3 and 30% of Grade 5 students met grade-level mathematics standards; the report states about two thirds had not achieved grade-level foundational skills	Bangla Grade 2-5 foundational numeracy kit	high / high
Telugu	ASER rural 2024: Andhra Pradesh Grade 3 at least subtraction 44.1% and Grade 5 division 36.2%; Telangana 30.9% and 25.2%	Telugu-English Grades 3-10 mastery bridge	high / medium
Indian Bengali	ASER rural West Bengal 2024: 40.9% of Grade 3 children could do at least subtraction and 35.0% of Grade 5 children could divide	Indian Bangla Grades 3-8 arithmetic-to-algebra catch-up	high / medium
Vietnamese	PISA 2022: 72% of represented 15-year-old students reached mathematics Level 2 or higher and 5% reached Levels 5-6	Precalculus-to-calculus-to-linear-algebra self-study spine	high / medium
Marathi	ASER rural Maharashtra 2024: 31.3% of Grade 3 children could do at least subtraction and 27.7% of Grade 5 children could divide	Marathi Grade 8-to-first-year-STEM bridge	high / medium
Indian Tamil	ASER rural Tamil Nadu 2024: 27.8% of Grade 3 children could do at least subtraction and 20.8% of Grade 5 children could divide	Tamil Grades 2-8 teacher-independent numeracy recovery	high / medium_high
Western Punjabi	ASER Pakistan 2023: 60.5% of rural Punjab Grade 5 children solved two-digit division; 2023 census counted 85,309,591 Punjabi mother-tongue residents in Punjab province	Shahmukhi Punjabi-to-Urdu/English secondary-to-UG bridge	high / medium
Javanese	No Javanese-specific large-scale mathematics assessment located; Indonesia PISA gives only a national proxy and BPS 2010 counted 68,044,660 age-5+ daily home users	Bilingual Javanese-to-Bahasa foundational/academic-register scaffold	medium / medium
Gujarati	ASER rural Gujarat 2024: 19.1% of Grade 3 children could do at least subtraction and 14.3% of Grade 5 children could divide	Gujarati Grades 2-6 diagnostic numeracy remediation	high / high

The Indian rows use rural state proxies and therefore do not turn one state percentage into a whole-language learning rate. Western Punjabi is the Pakistan/Shahmukhi profile, not Indian Gurmukhi

Punjabi. Indonesia's PISA evidence is not Javanese-specific. Those limits change the confidence label and product form; they do not justify replacing unknowns with zero.

*Table 5b sources. Bangladesh: Directorate of Primary Education (2023); Indian state proxies: ASER Centre (2025); Western Punjabi: ASER Pakistan Secretariat (2024) and Pakistan Bureau of Statistics (2023b); Indonesia and Viet Nam: Organisation for Economic Co-operation and Development (2023a, 2023b); Javanese home use: BPS-Statistics Indonesia (2010). Exact URLs and scope caveats are also retained in **population_mathematics_needs_register.csv**.*

7.4 Top 100 portfolio

The numbered Top 100 contains 100 natural-language interventions and 100 separately named localized outputs; no package occupies a cardinal position. Accessibility remains a separate, non-cardinal safeguard backlog and does not consume a linguistic position. The frozen FR-2 workload sums to 91,273,700 low, 407,304,900 base, and 1,866,457,100 high gross tokens. These compute totals are additive; the heterogeneous population measures are not summed as a count of unique people.

Portfolio lane	Interventions	Localized outputs	Base / optimistic / scarcity admissions	FR-2 base gross tokens
high_reach_underserved	35	35	12 / 13 / 10	142,556,715
regional_depth	49	49	21 / 21 / 7	199,579,401
small_population_prestige_domain_screen	16	16	7 / 6 / 3	65,168,784

7.5 Interlanguage and shared-core comparisons

No package replaces a natural-language row in the fail-closed cardinal portfolio. IL-PUNJABI, IL-HU, IL-NGUNI, and IL-SOTHO retain complete component joins and positive modeled savings, but their upstream population unions are not admissible. Their displayed population values are diagnostic component subtotals only: they are nonadditive, noncardinal, and not unique-person, family-level, or

constructed-bridge comprehension reach. Base reuse is an unobserved sensitivity, not an empirical production saving.

Table 6a. Shared-core package sensitivity models with complete component joins.

Architecture	Named outputs	Profiles	Diagnostic component subtotal (noncardinal)	Independent base tokens	Shared-core base sensitivity	Modeled saving sensitivity
IL-HU	2	ur-Arab-PK;ur-Arab-IN	72,975,069	8,146,098	6,109,574	2,036,524
IL-NGUNI	4	zu-Latn-ZA;xh-Latn-ZA;ss-Latn-ZA;nr-Latn-ZA	27,137,226	16,292,196	10,182,623	6,109,573
IL-PUNJABI	2	pa-Guru-IN;pnb-Arab-PK	120,059,639	8,146,098	6,109,574	2,036,524
IL-SOTHO	3	nso-Latn-ZA;tn-Latn-ZA;st-Latn-ZA	15,624,006	12,219,147	8,146,098	4,073,049

The other 11 package proposals are excluded because planned named-output coverage, exact population joins, constituent score joins, or interval reconciliation is incomplete. Constructed Interslavic has a zero registered cardinal lower bound because no exact sustained mathematical-reading population result is available; that is not a substantive estimate of zero mathematical usefulness. Mathematical formalism can reduce prose dependence for expert readers, while novice word problems and explanations remain language-heavy. The paper therefore treats expertise and formal density as a stage-specific interlanguage prior to be measured, not as a blanket family multiplier or a reason to dismiss the architecture. Reuse fractions remain modeled sensitivities rather than observed production savings.

Top-100 exact-profile crosswalk

The ordered Top 100 was joined to the registered interlanguage matrix without fuzzy family matching. Sixteen rows have exact target-profile matches; four more match an exact language/script cell whose named territory contains the target country; one Belarusian-Interslavic relation remains hypothesis-only because its territory is unresolved; and 79 rows have no exact current matrix relation. The 20 exact or

country-compatible rows can therefore enter shared-core engineering analysis, but none receives additional readership automatically. All 100 current cross-language demographic reach credits remain zero, and every row remains non-rankable on interlanguage evidence alone.

Table 6b. Exact Top-100-to-interlanguage overlap-control crosswalk.

Architecture	Relation class	Top-100 positions and exact targets
IL-IDMS	exact profile	1 Bahasa Indonesia (id-Latn-ID)
IL-PUNJABI	exact profile	8 Western Punjabi (pnb-Arab-PK); 18 Eastern Punjabi (pa-Guru-IN)
IL-HU	exact profile	14 Indian Urdu (ur-Arab-IN); 24 Pakistani Urdu (ur-Arab-PK)
IL-TURKIC	exact profile	17 Uzbekistan Uzbek (uz-Latn-UZ); 33 Turkmen (tk-Latn-TM); 35 Kazakh (kk-Cyrl-KZ); 42 Kyrgyz (ky-Cyrl-KG)
IL-PDT	exact profile	23 Dari (prs-Arab-AF); 38 Tajik (tg-Cyrl-TJ)
IL-NGUNI	exact profile	31 isiZulu (zu-Latn-ZA); 37 isiXhosa (xh-Latn-ZA); 75 isiNdebele (nr-Latn-ZA)
IL-MANDING	exact profile	41 Mali Bamanankan (bm-Latn-ML)
IL-SOTHO	exact profile	47 Sepedi (nso-Latn-ZA)
IL-NGUNI; IL-SOTHO; IL-GERM	exact language/script plus named country	68 siSwati (ss-Latn-ZA); 49 Setswana (tn-Latn-ZA); 50 Sesotho (st-Latn-ZA); 84 Pennsylvania German (pdc-Latn-US)
IL-ISV	hypothesis only	57 official-standard Belarusian (be-Cyrl-BY); territory unresolved

The crosswalk separates direct localized coverage, possible reuse of source alignment or terminology work, and additional people reached across languages. Shared-core reuse is plausible for the 20 matched rows but unquantified because no empirical token-savings coefficient exists. Indonesian is the limiting control: its exact Open Logic component is complete at 722/722 units, so its forward deficit is $D=0$. It can seed later adaptations, but it is existing own-language coverage and supplies no automatic Malaysian, Bruneian, Javanese, Sundanese, or other cross-language reach. The remaining 79 rows are unmapped under the current exact matrix; that is not evidence that interlanguage engineering is impossible.

7.6 Fixed-source OpenStax and Open Logic compute comparators

FR-2 at D3 (210 editable units; 120,083 measured source alpha tokens) remains the common language-priority compute comparator. The older territory-proxy rule also groups the 100 rows into fixed OpenStax scenarios. These are reproducible workload counterfactuals, not claims that all populations need Formal Reasoning Core first or that the listed OpenStax bundle is locally missing. Actual commissioning follows Table 5b and `top100_needs_assignment_v2.csv`: all 100 rows now carry an explicit first-product or bounded-audit assignment, while the 90 rows outside the directly audited Top 10 retain lower-confidence territory-proxy or audit-required status rather than fabricated language-specific findings.

Table 7. Fixed-source OpenStax comparator allocations for the Top 100.

Next portfolio	Depth	Interventions	Outputs	Low tokens	Base tokens	High tokens
MV-1	D2	36	36	39,137,220	156,816,432	670,681,044
MV-1	D3	43	43	69,334,447	235,382,774	906,173,443
SB-1	D3	21	21	62,944,707	238,856,625	989,359,497

Table 7 costs only the 100 cardinal natural-language interventions under fixed-source comparator scenarios. Architecture sensitivities remain in Table 6a and are not transferred into OpenStax compute or cardinal output totals. These totals must not be relabeled as the needs-optimal curriculum.

7.7 Sensitivity and negative controls

The informative rank range spans the base, optimistic, and scarcity views. The conservative zero floor is disclosed separately and does not manufacture an arbitrary strict order. Richly served controls remain available for non-overlap, subgroup, and accessibility tests but are not generic translation priorities. Unresolved profiles and D0 profile/population mismatches are exclusions, not evidence that their communities are already served.

Table 8. Richly served negative controls and profile exclusions.

Target	Profile	Territory	Gross base population	Measure	Year	Source ID	Classification
Polish	pl-Latn-PL	Poland	37,868,618	census_languages_used_at_home_multiple_respon se_persons	2021- 03-31	GRG-S006	richly_served_negative_control
German	de-Latn	Poland	216,342	census_languages_used_at_home_multiple_respon se_persons	2021- 03-31	GRG-S006	richly_served_negative_control
Ukrainian	uk-Cyrl-UA	Poland	55,104	census_languages_used_at_home_multiple_respon se_persons	2021- 03-31	GRG-S006	richly_served_negative_control
Czech	cs-Latn-CZ	Poland	5,328	census_languages_used_at_home_multiple_respon se_persons	2021- 03-31	GRG-S006	richly_served_negative_control

Table 8 also preserves 5 non-control exclusions: three unresolved localized-output profiles and two D0 profile/population mismatches. They are not counted as negative controls and must not be silently converted into recommendations.

The complete machine-readable tables preserve exact population measures, years, observation IDs, source IDs, factor statuses, compute scenarios, package exclusions, and tie-aware ranks.

8. Grant and implementation program

The recommended program is staged by reusable infrastructure and diagnosed learner need rather than by isolated PDFs or one universal book assignment.

1. **Open source and production eligibility.** Pin legally reusable editable Open Logic, OpenStax, and other open units; preserve formula identity, semantic structure, examples, exercises, public answers, attribution, tests, and deterministic builds. A source that cannot be translated, corrected, redistributed, printed, and preserved at zero learner price is not a program source.
2. **Population-specific needs audit.** For each exact language/variety/script/ territory profile, distinguish foundational numeracy, arithmetic-to-algebra recovery, bilingual academic-register transfer, secondary-to-tertiary bridging, coherent undergraduate sequencing, and advanced/reference scarcity. Language rank and curriculum rank are separate decisions.

3. **High-reach commissioning wave.** Start the corrected Top 10 with the first products in Table 5b - not ten copies of the same course. Continue or bridge existing work rather than duplicating it: Indonesian commissioning starts from the existing 40-course program and closes only proven gaps; Bangla, Tamil, and Gujarati begin lower in the prerequisite ladder; Vietnam begins higher.
4. **Shared-core and interlanguage wave.** Where related varieties support semantic, terminology, notation, or QA reuse, maintain one aligned core and release named localized outputs. In advanced, symbol-dense mathematics, instrument receptive interlanguage comprehension separately by expertise and task instead of assuming either blanket family coverage or blanket uselessness.
5. **Regional-depth and prestige-domain wave.** Fund smaller exact profiles whose scarcity, vitality, regional function, or serious-mathematics public function would be erased by population ranking. Japan's advanced algebraic-geometry lane belongs here as a content-gap intervention, not as generic school-algebra remediation.
6. **Teacher-independent access baseline.** Every learner package includes placement, prerequisite routing, worked examples, exercises, complete public answers, misconception explanations, semantic HTML/MathML, reflowable EPUB, checked screen and print PDFs, local fonts/assets, and a one-download offline bundle. Classroom guides and signed/audio/Braille derivatives extend the same source; they do not replace the independent learner edition.
7. **Maintenance and evaluation.** Preserve source hashes, terminology decisions, formula/structure checks, build evidence, model-audit results, corrections, and privacy-preserving usage indicators. New comprehension or usage evidence updates the next portfolio without making human availability a release prerequisite.

Table 9. Grant-scale fixed-source inference comparators.

Scope	Workflow	Cardinal interventions	Modeled outputs	Task bundle	Gross tokens	API-equivalent USD
Top-10 first-wave comparison	Low	10	10	10 x FR-2/D3	9,127,370	\$51.90
Top-10 first-wave comparison	Base	10	10	10 x FR-2/D3	40,730,490	\$167.70

Scope	Workflow	Cardinal interventions	Modeled outputs	Task bundle	Gross tokens	API-equivalent USD
Top-10 first-wave comparison	High	10	10	10 x FR-2/D3	186,645,710	\$654.30
Headline Top 100	Low	100	100	100 x FR-2/D3	91,273,700	\$519.00
Headline Top 100	Base	100	100	100 x FR-2/D3	407,304,900	\$1,677.00
Headline Top 100	High	100	100	100 x FR-2/D3	1,866,457,100	\$6,543.00
Top 100 plus fixed-source OpenStax comparator	Low	100	200	FR-2 plus 36 x MV-1/D2, 43 x MV-1/D3, 21 x SB-1/D3	262,690,074	\$1,963.61
Top 100 plus fixed-source OpenStax comparator	Base	100	200	FR-2 plus 36 x MV-1/D2, 43 x MV-1/D3, 21 x SB-1/D3	1,038,360,731	\$5,978.52
Top 100 plus fixed-source OpenStax comparator	High	100	200	FR-2 plus 36 x MV-1/D2, 43 x MV-1/D3, 21 x SB-1/D3	4,432,671,084	\$22,397.84
All 134 eligible natural-language interventions	Low	134	137	137 x FR-2/D3	125,044,969	\$711.03
All 134 eligible natural-language interventions	Base	134	137	137 x FR-2/D3	558,007,713	\$2,297.49
All 134 eligible natural-language interventions	High	134	137	137 x FR-2/D3	2,557,046,227	\$8,963.91

Note. Table 9 is deliberately a source-fixed compute comparator, not the budget for the needs-optimal curriculum. All 100 rows carry an explicit first-product or bounded- audit assignment; the first 10 use the directly audited needs layer in Table 5b, while the other 90 preserve lower-confidence territory-proxy or audit-required status rather than being represented as equally strong language-specific findings. Three eligible interventions each require two explicitly named localized outputs, so 134 interventions produce 137 modeled outputs in the full eligible set. A grant budget should replace each comparator with the exact selected source package and its measured token, figure, formula, exercise, accessibility, and maintenance burden. Dollar values use one convention throughout: each bundle's frozen two-decimal per-output planning cost is multiplied by its modeled output count and the bundle groups are summed. They are not aggregate-token repricing values. The planning comparator does not model a cache-write surcharge. Separately, the bounded empirical 33-root receipt reports its cache-write field as exactly zero; that observation is not projected beyond the receipt's scope. Non-API program costs are excluded, not zero.

Budget requests should separate inference, edition, and delivery costs. A low API bill does not imply low all-in cost where source acquisition, terminology, typography, accessibility, print/offline distribution, or maintenance dominate. Conversely, openly licensed semantic infrastructure is the condition that makes cumulative reuse, zero-price learner access, correction, and future-language expansion possible.

9. Limitations

Population sources differ in year, age universe, language question, and territorial coverage. Exact counts are exact only to their printed measure; they are not all current L1-speaker estimates. Many censuses measure ethnicity rather than language, speaking rather than reading, or household use rather than individual ability.

Academic-lingua-franca comfort is the largest unresolved factor. Official status, school exposure, or self-reported subsidiary-language knowledge does not prove comfortable advanced reading. Wide ranges are therefore a feature, not an error.

Scarcity proxies can miss unindexed, offline, paywalled, or locally hosted resources, and can overcount research works that are not teaching material. Model benchmarks are general-domain and may not represent specialist mathematics or the target variety.

The compute model is reproducible but scenario-based. It is calibrated to exact local source tokens and planned workflow stages; its coefficients are not fitted to the empirical 33-root cached/fresh/output boundary, and the broader 6,726-thread closure lacks a historical category split. API-equivalent prices are date- and model-specific and should not be treated as weekly subscription accounting or complete program cost.

Interlanguage evidence is especially sparse, but the evidentiary gap must be named correctly. A zero in the cardinal register means **no demographic reach has yet been credited**; it is not a substantive estimate that the surface has zero mathematical utility. Short-task comprehension, linguistic relatedness, or a family label cannot alone establish sustained mathematical usability, while symbol density, prior subject knowledge, and the reader's ability to reconstruct formal structure may make advanced mathematics unusually receptive to partial linguistic comprehension. The present portfolio therefore withholds unmeasured population multiplication, credits shared- core production savings

where auditable, and treats expertise-by-formal-density comprehension as a high-priority measurement program.

Finally, access is not learning, but this distinction does not reduce the product to a bare text or assume that a teacher must supply the rest. The ranking identifies where an open, exercise-rich, answer-bearing, offline-capable course can remove a barrier. Learning-adjusted causal claims still require a defined package, exposure, and outcome measure; access claims require proof that the usable package exists and can be obtained.

10. Conclusion

Efficient use of AI compute for educational access is a portfolio-design problem, not a speaker-count sort and not a race to maximize translated file totals. The central unit must be an exact production profile - language or variety, script, orthography, territory, register, and curriculum - joined to a dated population observation and an explicit non-overlap model. The revised analysis also makes a second distinction that the first paper missed: **the population that should be served and the mathematics it most needs are two different optimization questions.**

On the equal ex-ante comparison used for other natural-language candidates, Bahasa Indonesia ranks first. Its official 2022 BPS measure counts 248,501,794 people aged five or older reported able to understand spoken Indonesian and produce Indonesian words intelligible to another person; this is a functional-language ceiling, not a literacy or academic-readiness count. Including it corrects a classification error: completed Indonesian work made the language more feasible and reduced the next marginal cost, but the earlier model treated that evidence as a reason to omit the population opportunity. Bangladesh Bangla, Telugu, Indian Bengali, Vietnamese, Marathi, Indian Tamil, Western Punjabi, Javanese, and Gujarati complete the corrected Top 10 under the disclosed alternating

base/optimistic/scarcity rule. The list records a reproducible allocation view rather than pretending that one uncertain scalar identifies global welfare.

Those ten populations should not receive ten identical first books. For Indonesian, the first operation is no longer another book title: it is an exact needs-to-program crosswalk across the existing 40-course curriculum, whose public overlay records 27 published roles and 13 production roles, followed by closure of only the unfinished prerequisite, solution, delivery, or accessibility components. Its page evidence is reported as distinct teaching-package, selected-corpus, rendered-universe, and public-artifact totals rather than one false final translated-page count. The other first-wave packages are foundational Grade 2-5 numeracy for Bangladesh Bangla; arithmetic-to-algebra catch-up for Telugu, Indian Bengali, Tamil, and Gujarati; a secondary-to-tertiary bridge for Marathi and Western Punjabi; a coherent precalculus, calculus, linear-algebra, and proof spine for Vietnamese; and a bilingual Javanese-to-Bahasa academic-register scaffold whose outcome evidence is explicitly weaker. Japanese generic school algebra is deprioritized because national achievement and textbook supply are strong, while advanced proof, abstract algebra, commutative algebra, and algebraic geometry in Japanese remain a legitimate high-level scarcity and prestige-domain program. “What is missing?” must therefore be audited at the population, territory, stage, and subject level before compute is committed.

The 210-unit Formal Reasoning Core and the fixed OpenStax bundles remain useful because they make token comparisons reproducible, not because they are universal curricular prescriptions. Producing the 100 selected FR-2 comparison editions is estimated at 91.274 million gross tokens in the low workflow, 407.305 million in the base workflow, and 1.866 billion in the high workflow. The larger 100-language OpenStax comparison uses 36 MV-1/D2, 43 MV-1/D3, and 21 SB-1/D3 workloads. Actual grants should replace those fixed denominators with the measured source burden of the needs-audited package; all 100 Top-100 rows now carry a territory- and stage-specific first-product or bounded- audit assignment, with confidence and caveats explicit. The Indonesian implementation also shows why source-length estimates cannot be presented as observed program cost: 33 user-visible roots recorded 83.639 billion

total tokens with 97.714% of gross input cached, while the broader 6,726-thread closure recorded 10.253 trillion cumulative total tokens without a historical component split. A narrower nine-goal ledger attributes 88.493 million pursuit-accounting tokens to named phases. These boundaries are not added or priced; they motivate instrumentation that preserves scope, cached and fresh input, output, reasoning-subset, retries, and workflow stage for future grants.

Interlanguage work has two potentially large mathematical advantages. First, a shared semantic/terminology core can reduce production compute while preserving separately named localized editions; Punjabi, dual-territory Urdu, Nguni, and Sotho architectures show modeled base savings of 25%-37.5%. Second, advanced mathematics may permit more receptive cross-language or constructed-bridge reading than ordinary prose because notation, formal structure, diagrams, examples, and prior disciplinary knowledge carry part of the meaning. The effect should be strongest for expert readers and symbol-dense material and weakest for foundational explanation and word problems. Constructed Interslavic therefore receives no **unmeasured demographic multiplier** from a short cloze result, but neither Interslavic nor interlanguages generally are rated useless. They receive an explicit mathematics-specific research design: compare comprehension, error detection, and problem solving across expertise levels and prose/formal-density bands, then incorporate measured reach without double counting readers already served by native-language editions.

The program's baseline artifact is not a streaming portal and not a teacher-dependent promise. It is a lawful open source, a mathematically capable model translation and semantic-audit pipeline, preserved exercises and public worked answers, and a package that can be downloaded once to a phone or shared device and used offline. Classroom adoption, print, MathML/HTML, reflowable EPUB, plain-language, audio/Braille, and signed- language products expand that baseline. The separate accessibility safeguard backlog keeps those nested access axes visible without pretending that their global population ceilings are additive.

The resulting grant strategy is concrete: fund reusable open and formula-preserving source infrastructure; commission the corrected Top 10 according to demonstrated content need; extend to a needs-audited Top 100; test shared-core and receptive interlanguage mathematics as separately instrumented workflows; and publish offline and accessible derivatives alongside the language edition. As better literacy, academic-comfort, comprehension, production, and usage evidence arrives, the same open model can be rerun without discarding prior editions or converting uncertainty into zero.

Model and contributor provenance

The research questions, program-design considerations, corrections, and publication direction were supplied by the commissioning researcher. Literature retrieval, evidence reconciliation, quantitative analysis, drafting, source-to-formula checks, document construction, and revision assistance were performed with **OpenAI Codex gpt-5.6-sol, Ultra**. This identification describes the computational production environment; it does not replace the authorship, licensing, or contributor credit of any source work. Every reused source retains its stated author, publisher, license, and attribution, and the machine-readable evidence register preserves the authority and limitation attached to each quantitative claim.

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Appendix A. Reproducibility and computational provenance

Research synthesis, deterministic modeling, drafting, and document production used **OpenAI Codex gpt-5.6-sol, Ultra**. This model identity does not replace the authorities in the evidence register. Exact source labels, measures, dates, locators, caveats, and hashes remain attached to the corresponding machine-readable rows.

The table identifies the principal current inputs and derived tables. The release `MANIFEST.sha256` records the complete final public-file inventory; the paper deliberately does not attempt to hash itself inside this self-referential table.

Artifact	Data rows	Bytes	SHA-256
population_observations_master.csv	475	387,613	B43B4B3A48AC2BE647F4ED4446D566822F713351A7361D8F663D802A7990F3EF
population_source_register_public.csv	81	63,578	FE3AA4257E8FA8CC742DBA53FEF5414174975C93D6D4C262E034FFD2FDF3E4DF
candidate_interventions_master.csv	210	123,913	AEE2224E139FF346C459024A42BAC7C3F4B7D718A5CFF5D9691A59100762761B
natural_language_scores_v3.csv	106	138,598	096976FD45EB9B5EFC0D4D2C71730F861D51F6B18131AA56CD94876886A48BFD
candidate_expansion_scores.csv	37	50,444	B97383E20E9FAD2791BD4883082CF3985C66F08158670A466B9D82597BD43291
staging/interlanguage_bundle_model/interlanguage_intervention_matrix.csv	104	84,793	DD6BD32BCEC92FD2DE3111DBC584EBC7455E4C707241AF647B62F500318D8B33
staging/interlanguage_bundle_model/interlanguage_population_links.csv	113	49,082	8743D37ACB11EC2942EB3BE07AC9C4F884FD07DA8FCFF320E89B6775EC5A6DEB
interlanguage_package_scores.csv	135	88,114	5B943C3CDF2C2454F21ADD46B8897F1F61CFE32F06CA81E1561378D8C855BAE2
portfolio_compute_scenarios.csv	60	18,067	6FCFBE82135F0671BF30F03FE1F404FF8E9E9C054992A4445AA8ACA0016A4961

Artifact	Data rows	Bytes	SHA-256
curriculum_units.csv	29	7,074	0F7A958D7C19F989A1C95F673456B3667ED23F41B64208100EA1CB5C3360E204
curriculum_portfolios.csv	13	3,404	2E4CC460C4CB343A092C5A8A4C11EC09EEF091A93807D983B74E258A698B5049
openstax_workload_measurements.csv	94	26,411	ECF9E83428CFCBA41600AE070F970A9685E2C5F7E6E2F2045C07B1DB319A86A2
portfolio_linguistic_candidates.csv	134	164,909	0B99C00FC987F57C893BA8B0B1B91843D2CAEC0852F383249983904BC1945A9B
TOP_10.csv	10	12,533	E7287B63C063E21200F0D0C12E3C9E12828222ABF365563B4A6CF1BD1582FC48
TOP_100.csv	100	117,712	BB5894B135A77FA7EC6FBD854477025D0BF3ABA1FE476EA0F5430D6A9DF3CDE3
table2_exact_gross_ceilings.csv	134	97,622	3E655674420B1D5D25E4584FD5D88134E86FE5EFFBEE53671A16588A55937BB0
table3_marginal_access_sensitivity_ranges.csv	134	125,306	8225CA538A409B27402F5A4830DB94B2155415A6F3BFAA61262DAF120A84E7D3
table4_top10.csv	10	10,687	7768FF21264C79D6F075D143B2C61F5E9FBA8D82BFC2572793369D47AA80BFD3
table5_top100_grouped_lanes.csv	3	5,333	C8A01350D97C0A24FD09332DC1D2E3A6849FAEB977BFD5C167DA9473EEF27348
table5_top100_members.csv	100	65,610	5EE9376D77C392FD0C628A8ED1CD3F9091477090EC1108FC0DA281833C6B92D9
table6_package_exclusions_noncardinal.csv	11	13,407	40D2BF237CC9FED66CD20FBA0C71ABEB6DCFCB32C6422DF6885858CA1A2AC282
table6a_architecture_sensitivity_noncardinal.csv	4	10,163	A6A5A9B26F82F91882CC5EB5E759F1065C745D64277E181E2C60890E5D804CD0
table7_curriculum_allocation_summary.csv	4	3,545	34C83D6F7A93E30C8AACA980EBFC7246719956D791EB1AFA548148DF95F7D376
table8_negative_controls_and_exclusions.csv	9	5,584	B7CC8F6DCA3BFE006FA1104E6E6B6ED8849D72B67D23E76178EE76C2A61A AFC5
population_mathematics_needs_register.csv	11	14,695	0E070BE5F4660EDFA6F47FA1F6CD6309F253FCCE8BF1B9197C0A8895C606634C

Artifact	Data rows	Bytes	SHA-256
top100_needs_assignment_v2.csv	100	175,234	A34A4C10F9CF84522A7CEB49A50EC59D727AD9C1086E025D0C09C4ECAAF33CBB0
indonesian_equal_basis_and_forward_allocation.csv	2	2,132	2B142F64FC663A5B7917CCD0A0E85F8D311CF53101A665A07FC0EB14705454EB
appendix_a_existing_work_reconciliation.csv	210	115,956	C6B7B53F1D13D6C24A42B37A0CA43D60D360EFCFD06D14BFB17AFBAAF7C65C5F
appendix_b_global_gap_map.csv	134	258,714	398EFBDF645F54E400F11D348201C5C7348011F70F69F0F70536F6BCA1FF3813
appendix_b_regional_gap_summary.csv	21	5,373	996DD1EA51C22962D445C8E5211296F57CDC8AE695A7A08FE06FC845D39E8F50
appendix_c_accessibility_safeguards.csv	11	14,637	43355270FD730CD6A36DBC265142DE07D1AFD9D7C618962877B4CDC2337AFA2C
appendix_d_top100_curriculum_mapping.csv	100	51,148	D64CC56ED243CC4D842531D73C3D0DA32AA4A3E35246F6012136D9BC8169889C
appendix_f_interlanguage_matrix_summary.csv	15	19,555	BA41972EC17E027A6C3DB1E589B8070525845A1DD9D9492061A0A42FAF6A5F1C
top100_interlanguage_overlap_crosswalk.csv	100	114,561	ED88A556E1055655E3635AAA64C793A684E1C655713BA4F86AD50B851CC1A380
TOP100_INTERLANGUAGE_CROSSWALK_METHOD_20260830.md	-	3,982	BF59C23B002F8ED7ECE9FDEC16CE624BED28D27ACA4404A5642404C48354AE8
TOP100_INTERLANGUAGE_CROSSWALK_VALIDATION_RECEIPT_20260830.md	-	2,489	1D8F4E0EE5F531D39059C86074A5A3334094FAF6BD15C3415B483D6F14311BDD
compute_token_audit_33_roots_20260830.json	-	6,620	1DAFEAC9EB161204DF470D247811F4E29415EE8AB3A297E412DBDD619CC0911C
INDONESIAN_TASK_TOKEN_AUDIT_20260830.md	-	9,863	6EA38C1684E3D2C9C1420DCE1C67A90DCCDBF6217D9503F2FA7D957CB87E5108
INDONESIAN_PROGRAM_COMPUTE_AND_PAGE_RECONCILIATION_PUBLIC_20260830.md	-	4,635	9472BE4077C60AF35C37B6615719B51425C1F555022F9DB55A96F6EF50B1A458
INDONESIAN_PUBLIC_PROGRAM_AUDIT_20260830.md	-	20,920	5C34C78E6E9BEF9E7B75E9179DD31CDE98995D70CEFF7DDD82DA98ECBFB6E0D

Appendix B. Full ordered Top 100

Every row is one exact named natural-language edition. Population values retain the source's own measure and date; they are not summed as unique people. *Base access* is a labelled sensitivity, not an observed harmed-population count. All 100 rows carry an explicit first-product or bounded-audit assignment. The Top 10 use directly audited primary or official evidence; the other 90 retain lower-confidence territory-proxy or audit-required status rather than fabricated language-specific findings.

Pos.	Intervention	Production profile	Territory	Gross base population	Base access	Portfolio lane	Needs-audited first product/status	Source ID
1	Bahasa Indonesia	id-Latn-ID	Indonesia	248,501,794	119,280,861	high_reach_underserved	Use the verified 27-published/13-production catalog to commission only an exact remaining course or format gap	PM-S021
2	Bangladesh Bangla	bn-Beng-BD	Bangladesh	165,323,060	65,302,609	high_reach_underserved	Bangla Grade 2-5 number sense, operations, place value, fractions, measurement and pattern kit with diagnostics and complete worked answers	ASSEC-S006
3	Telugu	te-Telu-IN	India	80,912,459	57,833,869	high_reach_underserved	Telugu-English Grades 3-10 mastery bridge with placement, error-specific explanations and full solutions	PM-S001
4	Indian Bengali	bn-Beng-IN	India	96,177,835	69,699,712	high_reach_underserved	Bangla Grades 3-8 arithmetic/fractions-to-algebra catch-up with placement tests, short mastery units and full answers	PM-S001
5	Vietnamese	vi-Latn-VN	Vietnam	89,000,000	42,777,849	high_reach_underserved	Vietnamese precalculus-to-calculus-to-linear-algebra self-study spine, reusing suitable open local modules before retranslating	ASSEC-S007
6	Marathi	mr-Deva-IN	India	82,801,140	50,316,309	high_reach_underserved	Marathi Grade 8-to-first-year-STEM bridge with embedded fractions/algebra diagnostic, functions, trigonometry, precalculus and proof introduction	PM-S001
7	Indian Tamil	ta-Taml-IN	India	68,888,839	48,387,709	high_reach_underserved	Tamil Grades 2-8 numeracy recovery course with placement, mastery units, full answers and worked misconception explanations	PM-S001

Pos.	Intervention	Production profile	Territory	Gross base population	Base access	Portfolio lane	Needs-audited first product/status	Source ID
8	Western Punjabi	pnb-Arab-PK	Pakistan	88,915,544	26,167,845	high_reach_underserved	Shahmukhi Punjabi-to-Urdu/English algebra, functions, trigonometry, precalculus and proof bridge	PM-S003
9	Javanese	jv-Latn-ID	Indonesia	68,044,660	32,661,437	high_reach_underserved	Oral/bilingual Javanese-to-Bahasa number-sense through prealgebra/algebra scaffold with audio and deliberate terminology transfer	PM-S005
10	Gujarati	gu-Gujr-IN	India	55,036,204	34,089,878	high_reach_underserved	Gujarati Grades 2-6 diagnostic remediation in place value, operations, fractions and proportional reasoning	PM-S001
11	Thai	th-Thai-TH	Thailand	64,080,191	29,188,527	high_reach_underserved	Audit local algebra/statistics and tertiary supply before choosing a secondary-data or undergraduate bridge.	TH-S001
12	Odia	or-Orya-IN	India	34,059,266	23,106,806	high_reach_underserved	Audit local algebra/statistics and tertiary supply before choosing a secondary-data or undergraduate bridge.	PM-S001
13	Kannada	kn-Knda-IN	India	43,506,272	31,535,081	high_reach_underserved	Audit local algebra/statistics and tertiary supply before choosing a secondary-data or undergraduate bridge.	PM-S001
14	Indian Urdu	ur-Arab-IN	India	50,725,762	30,600,888	high_reach_underserved	Audit local algebra/statistics and tertiary supply before choosing a secondary-data or undergraduate bridge.	PM-S001
15	Burmese	my-Mymr-MM	Myanmar	36,817,344	17,217,151	high_reach_underserved	Audit local secondary supply, then commission a complete algebra/functions/trigonometry bridge only where it is the binding gap.	ASSEC-S006
16	Malayalam	ml-Mlym-IN	India	34,776,533	23,946,869	high_reach_underserved	Audit local algebra/statistics and tertiary supply before choosing a secondary-data or undergraduate bridge.	PM-S001
17	Uzbekistan Uzbek Latin-script edition	uz-Latn-UZ	Uzbekistan	35,700,000	17,850,000	high_reach_underserved	Audit local algebra/statistics and tertiary supply before choosing a secondary-data or undergraduate bridge.	ALG-S018
18	Eastern Punjabi	pa-Guru-IN	India	31,144,095	17,886,376	high_reach_underserved	Audit local algebra/statistics and tertiary supply before choosing a secondary-data or undergraduate bridge.	PM-S001

Pos.	Intervention	Production profile	Territory	Gross base population	Base access	Portfolio lane	Needs-audited first product/status	Source ID
19	Sundanese	su-Latn-ID	Indonesia	32,412,752	15,558,121	high_reach_underserved	Audit local algebra/statistics and tertiary supply before choosing a secondary-data or undergraduate bridge.	PM-S005
20	Sindhi	sd-Arab-PK	Pakistan	34,401,564	10,124,380	high_reach_underserved	Diagnose foundational readiness first; commission a numeracy-to-algebra recovery path only if the local corpus audit confirms the gap.	PM-S003
21	Amharic	am-Ethi-ET	Ethiopia	21,634,396	6,540,078	high_reach_underserved	Diagnose foundational readiness first; commission a numeracy-to-algebra recovery path only if the local corpus audit confirms the gap.	PM-S020
22	Cebuano	ceb-Latn-PH	Philippines	20,697,364	10,190,347	high_reach_underserved	Audit local algebra/statistics and tertiary supply before choosing a secondary-data or undergraduate bridge.	ASSEC-S008
23	Dari	prs-Arab-AF	Afghanistan	30,083,114	5,605,988	high_reach_underserved	Diagnose foundational readiness first; commission a numeracy-to-algebra recovery path only if the local corpus audit confirms the gap.	ASSEC-S018
24	Pakistani Urdu	ur-Arab-PK	Pakistan	22,249,307	6,547,971	high_reach_underserved	Diagnose foundational readiness first; commission a numeracy-to-algebra recovery path only if the local corpus audit confirms the gap.	PM-S003
25	Assamese	as-Beng-IN	India	14,816,414	10,118,381	high_reach_underserved	Audit local algebra/statistics and tertiary supply before choosing a secondary-data or undergraduate bridge.	PM-S001
26	Saraiki	skr-Arab-PK	Pakistan	28,849,579	8,490,431	high_reach_underserved	Diagnose foundational readiness first; commission a numeracy-to-algebra recovery path only if the local corpus audit confirms the gap.	PM-S003
27	Sinhala	si-Sinh-LK	Sri Lanka	14,948,168	6,931,465	high_reach_underserved	Audit local secondary supply, then commission a complete algebra/functions/trigonometry bridge only where it is the binding gap.	ASSEC-S006
28	India Maithili Devanagari edition	mai-Deva-IN	India	13,353,347	7,644,770	high_reach_underserved	Audit local algebra/statistics and tertiary supply before choosing a secondary-data or undergraduate bridge.	PM-S001

Pos.	Intervention	Production profile	Territory	Gross base population	Base access	Portfolio lane	Needs-audited first product/status	Source ID
29	Afghanistan Pashto Arabic-script edition	ps-Arab-AF	Afghanistan	18,753,110	3,494,642	high_reach_underserved	Diagnose foundational readiness first; commission a numeracy-to-algebra recovery path only if the local corpus audit confirms the gap.	ASSEC-S018
30	Nepali	ne-Deva-NP	Nepal	13,084,457	5,502,014	high_reach_underserved	Diagnose foundational readiness first; commission a numeracy-to-algebra recovery path only if the local corpus audit confirms the gap.	PM-S004
31	isiZulu	zu-Latn-ZA	South Africa	14,613,202	6,659,967	high_reach_underserved	Audit local secondary supply, then commission a complete algebra/functions/trigonometry bridge only where it is the binding gap.	ZA-S001
32	Khmer	km-Khmr-KH	Cambodia	14,893,134	5,356,316	high_reach_underserved	Audit local secondary supply, then commission a complete algebra/functions/trigonometry bridge only where it is the binding gap.	ALG-S011
33	Turkmen	tk-Latn-TM	Turkmenistan	6,297,965	3,145,834	regional_depth	Audit local secondary supply, then commission a complete algebra/functions/trigonometry bridge only where it is the binding gap.	ALG-S009
34	Shona	sn-Latn-ZW	Zimbabwe	11,251,734	5,244,996	high_reach_underserved	Diagnose foundational readiness first; commission a numeracy-to-algebra recovery path only if the local corpus audit confirms the gap.	PM-S011
35	Kazakh	kk-Cyrl-KZ	Kazakhstan	13,380,107	3,345,027	high_reach_underserved	Audit local algebra/statistics and tertiary supply before choosing a secondary-data or undergraduate bridge.	ALG-S007
36	Lao	lo-Lao-LA	Laos	5,487,956	2,075,545	regional_depth	Diagnose foundational readiness first; commission a numeracy-to-algebra recovery path only if the local corpus audit confirms the gap.	ASSEC-S006
37	isiXhosa	xh-Latn-ZA	South Africa	9,786,928	4,460,393	regional_depth	Audit local secondary supply, then commission a complete algebra/functions/trigonometry bridge only where it is the binding gap.	ZA-S001
38	Tajik	tg-Cyrl-TJ	Tajikistan	10,394,100	2,598,525	high_reach_underserved	Audit local algebra/statistics and tertiary supply before choosing a secondary-data or undergraduate bridge.	ASSEC-S006

Pos.	Intervention	Production profile	Territory	Gross base population	Base access	Portfolio lane	Needs-audited first product/status	Source ID
39	Ethiopia Somali Latin-script edition	so-Latn-ET	Ethiopia	4,609,274	1,393,384	regional_depth	Diagnose foundational readiness first; commission a numeracy-to-algebra recovery path only if the local corpus audit confirms the gap.	PM-S020
40	Indonesia Madurese Latin-script edition	mad-Latn-ID	Indonesia	7,743,533	3,716,896	regional_depth	Audit local secondary supply, then commission a complete algebra/functions/trigonometry bridge only where it is the binding gap.	PM-S005
41	Mali Bamanankan Latin-script edition	bm-Latn-ML	Mali	9,551,561	1,701,611	regional_depth	Diagnose foundational readiness first; commission a numeracy-to-algebra recovery path only if the local corpus audit confirms the gap.	AFG-S008
42	Kyrgyz	ky-Cyrl-KG	Kyrgyzstan	5,527,168	1,381,792	regional_depth	Audit local secondary supply, then commission a complete algebra/functions/trigonometry bridge only where it is the binding gap.	ALG-S008
43	Ilocano	ilo-Latn-PH	Philippines	7,098,503	3,494,948	regional_depth	Audit local secondary supply, then commission a complete algebra/functions/trigonometry bridge only where it is the binding gap.	ASSEC-S008
44	Wolof	wo-Latn-SN	Senegal	8,786,892	2,212,539	regional_depth	Diagnose foundational readiness first; commission a numeracy-to-algebra recovery path only if the local corpus audit confirms the gap.	AFG-S009
45	Sri Lankan Tamil	ta-Taml-LK	Sri Lanka	3,297,390	1,529,000	regional_depth	Audit local secondary supply, then commission a complete algebra/functions/trigonometry bridge only where it is the binding gap.	ASSEC-S006
46	Hiligaynon	hil-Latn-PH	Philippines	6,073,883	2,990,476	regional_depth	Audit local secondary supply, then commission a complete algebra/functions/trigonometry bridge only where it is the binding gap.	ASSEC-S008
47	South Africa Sepedi Latin-script edition	nso-Latn-ZA	South Africa	5,972,255	2,721,855	regional_depth	Audit local secondary supply, then commission a complete algebra/functions/trigonometry bridge only where it is the binding gap.	ZA-S001

Pos.	Intervention	Production profile	Territory	Gross base population	Base access	Portfolio lane	Needs-audited first product/status	Source ID
48	Mongolian	mn-Cyrl-MN	Mongolia	3,051,962	1,505,228	regional_depth	Audit local secondary supply, then commission a complete algebra/functions/trigonometry bridge only where it is the binding gap.	ASSEC-S006
49	South Africa Setswana Latin-script edition	tn-Latn-ZA	South Africa	4,972,787	2,266,348	regional_depth	Audit local secondary supply, then commission a complete algebra/functions/trigonometry bridge only where it is the binding gap.	ZA-S001
50	South Africa Sesotho Latin-script edition	st-Latn-ZA	South Africa	4,678,964	2,132,438	regional_depth	Audit local secondary supply, then commission a complete algebra/functions/trigonometry bridge only where it is the binding gap.	ZA-S001
51	Maori	mi-Latn-NZ	New Zealand	213,849	53,462	regional_depth	Audit local secondary supply, then commission a complete algebra/functions/trigonometry bridge only where it is the binding gap.	PM-S014
52	Indonesia Minangkabau Latin-script edition	min-Latn-ID	Indonesia	4,232,226	2,031,468	regional_depth	Audit local secondary supply, then commission a complete algebra/functions/trigonometry bridge only where it is the binding gap.	PM-S005
53	Ethiopia Tigrinya Ethiopic-script edition	ti-Ethi-ET	Ethiopia	4,324,933	1,307,427	regional_depth	Diagnose foundational readiness first; commission a numeracy-to-algebra recovery path only if the local corpus audit confirms the gap.	PM-S020
54	Ireland Irish Latin-script daily-use edition	ga-Latn-IE	Ireland	71,968	17,992	small_population_prestige_domain_screen	Audit local secondary supply, then commission a complete algebra/functions/trigonometry bridge only where it is the binding gap.	PM-S016
55	Indonesia Banjar Latin-script edition	bjn-Latn-ID	Indonesia	3,651,626	1,752,780	regional_depth	Audit local secondary supply, then commission a complete algebra/functions/trigonometry bridge only where it is the binding gap.	PM-S005
56	Senegal Pulaar, official Latin orthography	fuc-Latn-SN	Senegal	4,314,509	1,086,393	regional_depth	Diagnose foundational readiness first; commission a numeracy-to-algebra recovery path only if the local corpus audit confirms the gap.	AFG-S009

Pos.	Intervention	Production profile	Territory	Gross base population	Base access	Portfolio lane	Needs-audited first product/status	Source ID
57	Official-standard Belarusian	be-Cyrl-BY	Poland	17,325	4,331	small_population_prestige_domain_screen	Audit local secondary supply, then commission a complete algebra/functions/trigonometry bridge only where it is the binding gap.	GRG-S006
58	Waray	war-Latn-PH	Philippines	2,694,135	1,326,457	regional_depth	Audit local secondary supply, then commission a complete algebra/functions/trigonometry bridge only where it is the binding gap.	ASSEC-S008
59	Ethiopia Sidama Latin-script edition	sid-Latn-ET	Ethiopia	2,981,471	901,299	regional_depth	Diagnose foundational readiness first; commission a numeracy-to-algebra recovery path only if the local corpus audit confirms the gap.	PM-S020
60	Standard Lithuanian	lt-Latn-LT	Poland	5,422	1,356	small_population_prestige_domain_screen	Audit local secondary supply, then commission a complete algebra/functions/trigonometry bridge only where it is the binding gap.	GRG-S006
61	Indonesia Sasak Latin-script edition	sas-Latn-ID	Indonesia	2,691,127	1,291,741	regional_depth	Audit local secondary supply, then commission a complete algebra/functions/trigonometry bridge only where it is the binding gap.	PM-S005
62	South Africa Xitsonga Latin-script edition	ts-Latn-ZA	South Africa	2,784,279	1,268,935	regional_depth	Audit local secondary supply, then commission a complete algebra/functions/trigonometry bridge only where it is the binding gap.	ZA-S001
63	Indonesia Acehnese Latin-script edition	ace-Latn-ID	Indonesia	2,550,055	1,224,026	regional_depth	Audit local secondary supply, then commission a complete algebra/functions/trigonometry bridge only where it is the binding gap.	PM-S005
64	Brahui	brh-Arab-PK	Pakistan	2,778,670	817,763	regional_depth	Diagnose foundational readiness first; commission a numeracy-to-algebra recovery path only if the local corpus audit confirms the gap.	PM-S003
65	Kirundi	rn-Latn-BI	Burundi	2,261,931	1,130,966	regional_depth	Diagnose foundational readiness first; commission a numeracy-to-algebra recovery path only if the local corpus audit confirms the gap.	AFG-S011

Pos.	Intervention	Production profile	Territory	Gross base population	Base access	Portfolio lane	Needs-audited first product/status	Source ID
66	Indonesia Betawi Latin-script edition	bew-Latn-ID	Indonesia	2,244,648	1,077,431	regional_depth	Audit local secondary supply, then commission a complete algebra/functions/trigonometry bridge only where it is the binding gap.	PM-S005
67	Maithili Nepal profile	mai-Deva-NP	Nepal	3,222,389	1,105,279	regional_depth	Diagnose foundational readiness first; commission a numeracy-to-algebra recovery path only if the local corpus audit confirms the gap.	PM-S004
68	South Africa siSwati Latin-script edition	ss-Latn-ZA	South Africa	1,692,719	771,457	regional_depth	Audit local secondary supply, then commission a complete algebra/functions/trigonometry bridge only where it is the binding gap.	ZA-S001
69	Zimbabwean Ndebele	nd-Latn-ZW	Zimbabwe	1,599,324	745,525	regional_depth	Diagnose foundational readiness first; commission a numeracy-to-algebra recovery path only if the local corpus audit confirms the gap.	PM-S011
70	Ethiopia Wolaytta Latin-script edition	wal-Latn-ET	Ethiopia	1,627,955	492,131	regional_depth	Diagnose foundational readiness first; commission a numeracy-to-algebra recovery path only if the local corpus audit confirms the gap.	PM-S020
71	South Africa Tshivenda Latin-script edition	ve-Latn-ZA	South Africa	1,480,565	674,768	regional_depth	Audit local secondary supply, then commission a complete algebra/functions/trigonometry bridge only where it is the binding gap.	ZA-S001
72	Bhojpuri Nepal profile	bho-Deva-NP	Nepal	1,820,795	627,264	regional_depth	Diagnose foundational readiness first; commission a numeracy-to-algebra recovery path only if the local corpus audit confirms the gap.	PM-S004
73	Mauritius Mauritian Creole Latin-script edition	mfe-Latn-MU	Mauritius	1,045,558	492,917	regional_depth	Audit local secondary supply, then commission a complete algebra/functions/trigonometry bridge only where it is the binding gap.	PM-S013
74	Ethiopia Hadiyya Latin-script edition	hdy-Latn-ET	Ethiopia	1,253,894	379,052	regional_depth	Diagnose foundational readiness first; commission a numeracy-to-algebra recovery path only if the local corpus audit confirms the gap.	PM-S020

Pos.	Intervention	Production profile	Territory	Gross base population	Base access	Portfolio lane	Needs-audited first product/status	Source ID
75	South Africa isiNdebele Latin-script edition	nr-Latn-ZA	South Africa	1,044,377	475,975	regional_depth	Audit local secondary supply, then commission a complete algebra/functions/trigonometry bridge only where it is the binding gap.	ZA-S001
76	Standard Haitian Creole	ht-Latn-HT	United States	867,500	216,875	regional_depth	Audit local secondary supply, then commission a complete algebra/functions/trigonometry bridge only where it is the binding gap.	GRG-S001
77	Indonesia Nias Latin-script edition	nia-Latn-ID	Indonesia	747,168	358,641	regional_depth	Audit local secondary supply, then commission a complete algebra/functions/trigonometry bridge only where it is the binding gap.	PM-S005
78	Avadhi Nepal profile	awa-Deva-NP	Nepal	864,276	276,568	regional_depth	Diagnose foundational readiness first; commission a numeracy-to-algebra recovery path only if the local corpus audit confirms the gap.	PM-S004
79	Zimbabwe Ndau Latin-script edition	ndc-Latn-ZW	Zimbabwe	372,607	173,691	regional_depth	Diagnose foundational readiness first; commission a numeracy-to-algebra recovery path only if the local corpus audit confirms the gap.	PM-S011
80	Silesian in ślabikōrzowy szrajbōnek	szl-Latn-PL	Poland	467,145	116,786	regional_depth	Audit local secondary supply, then commission a complete algebra/functions/trigonometry bridge only where it is the binding gap.	GRG-S006
81	Tetum Prasa	tet-Latn-TL	Timor-Leste	361,027	125,276	regional_depth	Diagnose foundational readiness first; commission a numeracy-to-algebra recovery path only if the local corpus audit confirms the gap.	ALG-S014
82	Senegal Mandinka (Màndienka), official Latin profile	mnk-Latn-SN	Senegal	465,052	117,100	regional_depth	Diagnose foundational readiness first; commission a numeracy-to-algebra recovery path only if the local corpus audit confirms the gap.	AFG-S009
83	YIVO Standard Yiddish edition for the explicitly named YIVO-reading audience	ydd-Hebr-US	United States	194,800	48,700	regional_depth	Audit local secondary supply, then commission a complete algebra/functions/trigonometry bridge only where it is the binding gap.	GRG-S001

Pos.	Intervention	Production profile	Territory	Gross base population	Base access	Portfolio lane	Needs-audited first product/status	Source ID
84	English-oriented Pennsylvania German spelling used by Vitt Du Deitsh Shvetza / Di Heilich Shrift / Ich Kann PA Deitsh Shreiva	pdn-Latn-US	United States	185,200	46,300	regional_depth	Audit local secondary supply, then commission a complete algebra/functions/trigonometry bridge only where it is the binding gap.	GRG-S001
85	Timor-Leste Makasai Latin-script edition	mkz-Latn-TL	Timor-Leste	123,840	42,972	regional_depth	Diagnose foundational readiness first; commission a numeracy-to-algebra recovery path only if the local corpus audit confirms the gap.	ALG-S014
86	Diné Bizaad in Navajo Nation school orthography	nv-Latn-US	United States	161,200	40,300	regional_depth	Audit local secondary supply, then commission a complete algebra/functions/trigonometry bridge only where it is the binding gap.	GRG-S001
87	Peru Ashaninka Latin-script edition	cni-Latn-PE	Peru	73,567	34,451	small_population_prestige_domain_screen	Diagnose foundational readiness first; commission a numeracy-to-algebra recovery path only if the local corpus audit confirms the gap.	PM-S007
88	New Zealand Samoan Latin-script edition	sm-Latn-NZ	New Zealand	110,541	27,635	regional_depth	Audit local secondary supply, then commission a complete algebra/functions/trigonometry bridge only where it is the binding gap.	PM-S014
89	Peru Awajun Latin-script edition	agr-Latn-PE	Peru	56,584	26,498	small_population_prestige_domain_screen	Diagnose foundational readiness first; commission a numeracy-to-algebra recovery path only if the local corpus audit confirms the gap.	PM-S007
90	School-standard Kashubian	csb-Latn-PL	Poland	89,198	22,300	small_population_prestige_domain_screen	Audit local secondary supply, then commission a complete algebra/functions/trigonometry bridge only where it is the binding gap.	GRG-S006
91	Timor-Leste Kemak Latin-script edition	kem-Latn-TL	Timor-Leste	68,995	23,941	small_population_prestige_domain_screen	Diagnose foundational readiness first; commission a numeracy-to-algebra recovery path only if the local corpus audit confirms the gap.	ALG-S014
92	Jamaican Creole in Cassidy/JLU orthography	jam-Latn-JM	United States	72,000	18,000	small_population_prestige_domain_screen	Audit local secondary supply, then commission a complete algebra/functions/trigonometry bridge only where it is the binding gap.	GRG-S001

Pos.	Intervention	Production profile	Territory	Gross base population	Base access	Portfolio lane	Needs-audited first product/status	Source ID
93	Timor-Leste Bunak Latin-script edition	bfu-Latn-TL	Timor-Leste	64,686	22,446	small_population_prestige_domain_screen	Diagnose foundational readiness first; commission a numeracy-to-algebra recovery path only if the local corpus audit confirms the gap.	ALG-S014
94	Timor-Leste Tokodede Latin-script edition	tkd-Latn-TL	Timor-Leste	46,784	16,234	small_population_prestige_domain_screen	Diagnose foundational readiness first; commission a numeracy-to-algebra recovery path only if the local corpus audit confirms the gap.	ALG-S014
95	Peru Shipibo-Konibo Latin-script edition	shp-Latn-PE	Peru	34,152	15,993	small_population_prestige_domain_screen	Diagnose foundational readiness first; commission a numeracy-to-algebra recovery path only if the local corpus audit confirms the gap.	PM-S007
96	Fataluku	ddg-Latn-TL	Timor-Leste	41,500	14,401	small_population_prestige_domain_screen	Diagnose foundational readiness first; commission a numeracy-to-algebra recovery path only if the local corpus audit confirms the gap.	ALG-S014
97	Standard 'Ōlelo Hawai'i with 'okina and kahakō	haw-Latn-US	United States	32,730	8,182	small_population_prestige_domain_screen	Audit local secondary supply, then commission a complete algebra/functions/trigonometry bridge only where it is the binding gap.	GRG-S001
98	Standard Marshallese under the 2010 Orthography Act	mh-Latn-MH	United States	27,160	6,790	small_population_prestige_domain_screen	Audit local secondary supply, then commission a complete algebra/functions/trigonometry bridge only where it is the binding gap.	GRG-S001
99	Timor-Leste Waima'a Latin-script edition	wmh-Latn-TL	Timor-Leste	21,227	7,366	small_population_prestige_domain_screen	Diagnose foundational readiness first; commission a numeracy-to-algebra recovery path only if the local corpus audit confirms the gap.	ALG-S014
100	Standard Central Alaskan Yup'ik	esu-Latn-US	United States	16,120	4,030	small_population_prestige_domain_screen	Audit local secondary supply, then commission a complete algebra/functions/trigonometry bridge only where it is the binding gap.	GRG-S001

Appendix C. Ordered accessibility safeguard portfolio

Only safeguard 1 has foundational priority. Items 2-11 retain the deterministic registered backlog order and are not evidence-based cardinal ranks. Every access-gain interval has a conservative zero floor or no defensible denominator. Token increments are FR-2 reference sensitivities, not observed usage, billing, or whole-programme cost.

Order	Safeguard	Barrier	Stratum	Source ceiling	Gain status	FR-2 base-token increment	Product	Nonadditivity	Population source
1	Semantic HTML with accessible math	visual_disability	India	771288	zero_to_nested_ceiling_only	610957	Born-accessible localized HTML; semantic headings tables language metadata MathML intent/structure skip links keyboard navigation and tested screen-reader reading order	ACCPop-VIS-INDIA-5-19-2011; all intervention edges in VIS-INDIA-5-19-2011	ACC-S004
2	Formula narration and synchronized audio	visual_disability	India	771288	zero_to_nested_ceiling_only	1425567	Deterministic language-specific symbol and formula narration synchronized with navigable text; diagrams require separate descriptions	ACCPop-VIS-INDIA-5-19-2011; all intervention edges in VIS-INDIA-5-19-2011	ACC-S004
3	English mathematical Braille derivative	braille_maths	United States and Canada		unbounded_unknown		Source-linked BRF using the jurisdictionally correct UEB technical or UEB-with-Nemeth policy; tactile/diagram alternatives; round-trip formula QA	visual impairment and screen-reader populations	ACC-S012
4	Localized British Sign Language instructional video and interactive glossary	signed_language	Scotland	117349	zero_to_language_ability_ceiling_only		Native BSL presentation with captions transcript downloadable low-bandwidth video glossary diagrams and source-aligned segment identifiers	none	ACC-S006
5	Logic figure and proof-tree remediation	visual_disability	India	771288	zero_to_nested_ceiling_only	346209	Programmatic proof structure with explicit subproof starts ends indentation rule names symbol pronunciation and linearized fallback	ACCPop-VIS-INDIA-5-19-2011; all intervention edges in VIS-INDIA-5-19-2011	ACC-S004
6	Mathematics figure remediation	visual_disability	India	771288	zero_to_nested_ceiling_only	610957	Structured data plus concise alt text and adjacent long descriptions; non-colour-only encodings; downloadable data where meaningful	ACCPop-VIS-INDIA-5-19-2011; all intervention edges in VIS-INDIA-5-19-2011	ACC-S004

Order	Safeguard	Barrier	Stratum	Source ceiling	Gain status	FR-2 base-token increment	Product	Nonadditivity	Population source
7	Localized New Zealand Sign Language instructional video and interactive glossary	signed_language	New Zealand	24680	zero_to_language_ability_ceiling_only		Native NZSL presentation with captions transcript downloadable low-bandwidth video glossary diagrams and source-aligned segment identifiers	none	ACC-S005
8	Offline HTML EPUB or SCORM package	connectivity	Global	2200000000	zero_to_global_ceiling_only	346209	Dependency-free static HTML plus conformant EPUB; local assets search navigation checksums and print profile; no mandatory account analytics CDN font or runtime requests	every language territory and learner cell; all other delivery-axis interventions	ACC-S002
9	Tagged searchable screen and print PDF	visual_disability	India	771288	zero_to_nested_ceiling_only	305479	Tagged reading order bookmarks language metadata embedded fonts searchable text black-and-white print profile and formula alternatives	ACCPop-VIS-INDIA-5-19-2011; all intervention edges in VIS-INDIA-5-19-2011	ACC-S004
10	Plain-language companion	cognitive_access_proxy	India	257755	zero_to_nested_proxy_ceiling_only	1934698	Aligned summaries definitions one-idea steps worked examples explicit goals consistent vocabulary and source links; preserve mathematical truth conditions	ACCPop-COG-INDIA-5-19-2011; AX-PLAIN edges	ACC-S004
11	Localized Peruvian Sign Language instructional video and interactive glossary	signed_language	Peru	10447	zero_to_childhood_language_ceiling_only		Native Peruvian Sign Language presentation with Spanish captions transcript downloadable low-bandwidth video glossary diagrams and source-aligned segment identifiers	none	ACC-S007

Appendix D. Existing-work reconciliation and deficit effect

D1. Exact current local baselines

The table below is the bounded successor-state census used to avoid commissioning a duplicate full Open Logic edition. It is separate from the 210-row candidate register: the single **covered** record is an explicit completed-baseline sentinel, while the other complete local editions remain baseline evidence rather than new ranked interventions.

Exact profile or architecture	Current bounded state	Marginal treatment	Controlling asset / SHA-256
zh-Hans-CN	Complete Open Logic; OpenStax Algebra and Trigonometry 2e 94/94; Calculus Volume 1 3/55 at the frozen state	No duplicate Open Logic commission; remaining OpenStax scope only	ACE-A006 / 89B03BB2854AE28DCBA29B73887FA20B46296DC9 4B8077C21CF90866439D331D
hi-Deva-IN	Complete, published Open Logic with public-byte readback	Baseline for Hindi; never credited to Urdu or other South Asian profiles	ACE-A007 / C155255517CA9B57B9556ABD18CADB8A4E31E527 25934F8A090C25D8C7D3B1C5
tr-TR	Complete Open Logic	Baseline for Turkish only; no automatic Turkic-family coverage	ACE-A008 / 8119903A18489AFEBEB18A1A787020869BB4F11E 32C008C8849D351392B77834
ar (Modern Standard Arabic formal edition)	Complete, published Open Logic with public-byte readback	Baseline formal Arabic coverage; dialect, Amazigh, Kurdish, Nubian, South Arabian, signed, and accessibility needs remain separate	ACE-A009 / 8F0DB97861BE0777A49359E865531AC2E6DBF42B 1BFA4D550A8817A000CC914B
fa-Arab-IR	Complete, published Iranian Persian Open Logic with public-byte readback	Baseline for Iranian Persian only; Dari, Tajik, Hazaragi, and other Iranic profiles remain separate	ACE-A009 / 8F0DB97861BE0777A49359E865531AC2E6DBF42B 1BFA4D550A8817A000CC914B
es	Complete local Open Logic QA state	Baseline for exact Spanish output; no pan-Romance reach	ACE-A010 / F2A940018EB87B959112B71EE8435D23C24469925 728CBFDD8AF15717B4AFD0
pt-BR	Complete local Open Logic QA state	Baseline for Brazilian Portuguese only; no European/African Portuguese or pan-Romance reach	ACE-A010 / F2A940018EB87B959112B71EE8435D23C24469925 728CBFDD8AF15717B4AFD0

Exact profile or architecture	Current bounded state	Marginal treatment	Controlling asset / SHA-256
id-ID	Complete Open Logic: 722/722 editable targets; 1,116-page public linked reader	Residual Open Logic deficit D = 0; no duplicate completion workload or reach	OpenLogic-id published release commit 34af65419e4c5c5580dae60a48454c485ddf504c; reader SHA-256 BF538D5E1994A7A7600703C9D24616696F77E43E9 312FB51078095FF0C963C0A
isv-Latn / isv-Cyrl programme	Seven accepted Open Logic units of 722; no finished reader	Unit progress only; no scalar token deficit or cross-language demographic reach inferred	ACE-A012 / 8515634D615773223074590F0DB4F2612BDF564B DDEC8E8015A6D6BB7815D3FB
Proposed Romance constructed surface	722 provisional files but zero canon-admitted units	Research/architecture hypothesis only; zero cardinal reach	ACE-A010 / F2A940018EB87B959112B71EE8435D23C24469925 728CBFDD8AF15717B4AFD0

The **id-ID** row is a completed Open Logic baseline, not a proxy for the whole Indonesian program. The pinned public overlay finds 27 of 40 course roles published and 13 in production. Its 26,031-page de-duplicated public-artifact reconstruction is distinct from the central ledger's 19,745 teaching-package pages, 20,763 selected-corpus working pages, and 27,705-page rendered universe; none is a final translated-page total. Consequently, Indonesian forward allocation begins with a course-by-course and format-by-format gap audit. It does not prescribe another Open Logic, Prealgebra, or Elementary Algebra translation and does not infer a new missing book merely from a population-level learning deficit.

D2. Candidate-register reconciliation states

State	Rows	Registered meaning	Deficit effect	Boundary
covered	1	Exact declared target scope is registered as delivered and verifiable.	D=0 only for the exact verified target × unit × format; any uncovered curriculum or format remains D=1.	Classification uses only existing_local_status and the declared scope in candidate_interventions_master.csv.
partial	18	At least one exact unit/component is registered, but the declared target scope is incomplete or inconsistent.	D=0 only on verified covered units/components and D=1 on the residual; no scalar is assigned without a common exact denominator.	Classification uses only existing_local_status and the declared scope in candidate_interventions_master.csv.

State	Rows	Registered meaning	Deficit effect	Boundary
researched	2	Research exists, but no delivered target coverage is registered.	Research without delivered target bytes does not reduce the content deficit; D=1 for the missing comparator.	Classification uses only existing_local_status and the declared scope in candidate_interventions_master.csv.
dormant	0	A prior item is registered as dormant and lacks current usable/build evidence.	A dormant item is not credited without verified usable current bytes and build identity; D=1 pending that registered evidence.	No rows in the current 210-row register meet this state; retained as an explicit fail-closed rule.
duplicated	0	The same usable target coverage is registered more than once; only a canonical copy may count.	The canonical item counts once; a duplicate adds no marginal coverage and produces no further reduction in D.	No rows in the current 210-row register meet this state; retained as an explicit fail-closed rule.
missing	189	No exact target coverage is registered in the bounded local census.	No registered target coverage: D=1 for the declared target $\times$ unit $\times$ format.	Classification uses only existing_local_status and the declared scope in candidate_interventions_master.csv.

Indonesian Open Logic is a complete 722/722 baseline: verified coverage is 722 units, the residual is zero, and D=0 for that exact corpus. It is covered rather than partial and cannot be counted again as forward translation gain. Interslavic has seven accepted units but no comparable accepted-source-token denominator, so this paper does not convert those local units into demographic bridge reach. All other partial rows remain component-level. The detailed 210-row reconciliation is machine-readable in [appendix_a_existing_work_reconciliation.csv](#).

Appendix E. Global regional, source, measure, and confidence gap map

Region	Subregion	Cardinal rows	Top100	Sources	Measures	High	Medium	Aggregation rule
Africa	Eastern Africa	1	1	1	derived_census_any_kirundi_read_and_written_persons_age_10_plus	1	0	not summed: measures, years, territories, and overlap universes may differ
Africa	Horn of Africa	6	6	1	census_mother_tongue_persons	6	0	not summed: measures, years, territories, and overlap universes may differ
Africa	Indian Ocean	1	1	1	language_usually_spoken_at_home_persons	1	0	not summed: measures, years, territories, and overlap universes may differ
Africa	Southern Africa	12	12	2	census_language_most_often_spoken_in_household_persons_age_1_plus census_mother_tongue_persons	12	0	not summed: measures, years, territories, and overlap universes may differ
Africa	Western Africa	4	4	2	census_mother_tongue_persons_age_3_plus census_principal_language_commonly_spoken_persons_age_3_plus	3	1	not summed: measures, years, territories, and overlap universes may differ
Americas	Arctic	1	0	1	acs_2017_2021_estimated_persons_age5plus_language_spoken_at_home	1	0	not summed: measures, years, territories, and overlap universes may differ
Americas	North America	13	2	1	acs_2017_2021_estimated_persons_age5plus_language_spoken_at_home	13	0	not summed: measures, years, territories, and overlap universes may differ
Americas	North America (Caribbean and Central American diaspora)	1	0	1	acs_2017_2021_estimated_persons_age5plus_language_spoken_at_home	1	0	not summed: measures, years, territories, and overlap universes may differ
Americas	North America (Caribbean diaspora)	3	2	1	acs_2017_2021_estimated_persons_age5plus_language_spoken_at_home	3	0	not summed: measures, years, territories, and overlap universes may differ
Americas	North America (Central American Indigenous diaspora)	4	0	1	acs_2017_2021_estimated_persons_age5plus_language_spoken_at_home	4	0	not summed: measures, years, territories, and overlap universes may differ

Region	Subregion	Cardinal rows	Top100	Sources	Measures	High	Medium	Aggregation rule
Americas	North America (European nonterritorial diaspora)	2	2	1	acs_2017_2021_estimated_persons_age5plus_language_spoken_at_home	2	0	not summed: measures, years, territories, and overlap universes may differ
Americas	North America (Pacific diaspora)	8	2	1	acs_2017_2021_estimated_persons_age5plus_language_spoken_at_home	8	0	not summed: measures, years, territories, and overlap universes may differ
Americas	South America	3	3	1	census_language_learned_in_childhood_age3plus_persons	3	0	not summed: measures, years, territories, and overlap universes may differ
Asia	Central Asia	5	5	5	census_mother_tongue_persons census_native_language_of_own_nationality_persons census_native_language_persons derived_cldr_territory_functional_language_users preliminary_census_mother_tongue_persons_rounded	4	1	not summed: measures, years, territories, and overlap universes may differ
Asia	East Asia	1	1	1	derived_cldr_territory_functional_language_users	0	1	not summed: measures, years, territories, and overlap universes may differ
Asia	South Asia	26	26	5	census_mother_tongue_persons census_same_name_mother_tongue_component_persons derived_2020_total_speakers_from_official_secondary_share_and_world_bank_denominator derived_cldr_territory_functional_language_users	21	5	not summed: measures, years, territories, and overlap universes may differ
Asia	Southeast Asia	26	26	8	census_mother_tongue_persons CLEAR_normalized_main_language_weighted_persons_from_IPUMS_2010_census_extract daily_language_at_home_age5plus_persons derived_cldr_territory_functional_language_users derived_union_of_mutually_exclusive_census_strata_persons official_long_form_census_able_to_speak_indonesian_persons_age5plus peer_reviewed_published_lower_bound_speakers	23	3	not summed: measures, years, territories, and overlap universes may differ
Europe	Central Europe	4	4	1	census_languages_used_at_home_multiple_response_persons	4	0	not summed: measures, years, territories, and overlap universes may differ

Region	Subregion	Cardinal rows	Top100	Sources	Measures	High	Medium	Aggregation rule
Europe	Northern Europe	1	1	1	self_reported_daily_use_outside_education_persons	1	0	not summed: measures, years, territories, and overlap universes may differ
Oceania	Australia	10	0	4	census_language_used_at_home_persons	10	0	not summed: measures, years, territories, and overlap universes may differ
Oceania	Polynesia	2	2	1	census_languages_spoken_multiple_response_persons	2	0	not summed: measures, years, territories, and overlap universes may differ

The global detail table has one row for each of the 134 cardinal natural-language interventions and an exact join to one population observation and its registered source. The source/measure/confidence table preserves source-level groupings. Counts in this appendix describe records and coverage strata; they are not summed population claims.

Appendix F. Curriculum portfolios, adaptation depths, and needs assignment

F1. Exact curriculum portfolios

ID	Portfolio	Project	Exact content	Units	Source tokens	Preferred depth	Prerequisite	Sources	Caveat
MV-1	Minimum viable quantitative curriculum	OpenStax	Algebra and Trigonometry 2e	94	-	D2	-	ACE-E002;ACE-E003	Whole book retained because later statistics physics and calculus reuse its sequence and vocabulary.
SB-1	Secondary data bridge	OpenStax	MV-1 plus Introductory Statistics 2e	196	-	D2	MV-1	ACE-E002;ACE-E003	Statistics chapters are not fully independent; partial extracts must be labeled as packets.
SB-2	Secondary STEM bridge	OpenStax	SB-1 plus College Physics 2e	479	-	D2_then_D3	SB-1	ACE-E002;ACE-E003	College Physics is algebra based and does not require calculus.
UG-1	Undergraduate STEM breadth	OpenStax	SB-2 plus Chemistry 2e and Biology 2e	887	-	D2	SB-2	ACE-E002;ACE-E003	Terminology figure and alt-text burden rises substantially.
UG-2	Complete undergraduate STEM core	OpenStax	UG-1 plus Calculus Volumes 1 through 3	1050	-	D2_then_D3	UG-1	ACE-E002;ACE-E003	Calculus volumes must be sequenced 1 then 2 then 3.
FR-1	Formal-reasoning foundation	Open Logic Project	methods plus sets-functions-relations plus propositional	87	50330	D2	-	ACE-A013;ACE-E004;ACE-E005	Compact mathematical-reasoning route.
FR-2	Formal-reasoning core	Open Logic Project	FR-1 plus first-order-logic	210	120083	D2	FR-1	ACE-A013;ACE-E004;ACE-E005	Preferred main Open Logic educational intervention.
FR-3	Logic and computation bridge	Open Logic Project	FR-2 plus computability plus turing-machines	276	157048	D2	FR-2	ACE-A013;ACE-E004	Suitable as a logic/computer-science bridge.
FR-4	Advanced formal-reasoning portfolio	Open Logic Project	full canonical reader	642	318916	D1_or_D2	FR-2	ACE-A013;ACE-E004	Select by official remix rather than repository order.
FR-5	Complete archival corpus	Open Logic Project	canonical plus supplements	722	367220	D1_unless_selected	FR-4	ACE-A013;ACE-E004	Not automatically a teachable sequence.

ID	Portfolio	Project	Exact content	Units	Source tokens	Preferred depth	Prerequisite	Sources	Caveat
AX-1	Core accessibility package	Cross-project	semantic HTML or MathML plus screen PDF print PDF and offline package	-	-	parallel_to_D2	-	ACE-E003;ACE-E005	Apply to every selected language and curriculum tier rather than deferring.
AX-2	Enhanced nonvisual access	Cross-project	formula narration diagram descriptions and audio or TTS-ready text	-	-	separate_derivative	AX-1	ACE-E003;ACE-E005	Requires deterministic mathematical and chemistry reading conventions.
AX-3	Plain-language companion	Cross-project	definitions summaries and worked-example explanations	-	-	separate_derivative	AX-1	ACE-E003;ACE-E005	Supplement only; never silently replaces the source-faithful edition.

F2. Exact adaptation depths

ID	Depth	Included components	Low	Base	High	Status	Caveat
D0	Terminology and structure	TOC metadata learning objectives key-term graph notation registry attribution	0.05	0.075	0.10	infrastructure	Not an educational edition.
D1	Reading edition	Expository narrative definitions summaries captions and a small worked-example set	0.30	0.375	0.45	reference_or_early_reach	Bulk assessment and practice architecture omitted.
D2	Course-ready core	Complete exposition definitions worked examples checkpoints representative exercises and answers figures and alt text	0.60	0.70	0.80	minimum_course_intervention	Default first educational production depth.
D3	Complete public edition	Every public module component exercise public answer caption glossary reference and attribution	1.00	1.00	1.00	complete_public_source	Does not include restricted instructor-only ancillaries.
D4	Local pedagogical adaptation	D3 plus locally appropriate data units software instructions examples and explanatory scaffolding	1.20	1.40	1.60	localized_complete	Separate from source-faithful translation and requires explicit adaptation labeling.

F3. Legacy fixed-source Top100 workload mapping

This mapping preserves the older uniform curriculum assignment solely as a reproducible compute sensitivity. It is **not** a claim that the named book is the first missing product in every population. The distribution is $MV-1/D2 = 36$, $MV-1/D3 = 43$, and $SB-1/D3 = 21$. SB-1 means **MV-1 plus Introductory Statistics 2e**.

Pos.	ID	Target	Profile	Lane	First comparator	First depth	Next comparator	Next depth	Next exact content	Population source
1	NAT-121	Bahasa Indonesia	id-Latn-ID	high_reach_underserved	FR-2	D2	SB-1	D3	MV-1 plus Introductory Statistics 2e	PM-S021
2	NAT-001	Bangladesh Bangla	bn-Beng-BD	high_reach_underserved	FR-2	D2	MV-1	D3	Algebra and Trigonometry 2e	ASSEC-S006
3	NAT-003	Telugu	te-Telu-IN	high_reach_underserved	FR-2	D2	SB-1	D3	MV-1 plus Introductory Statistics 2e	PM-S001
4	NAT-002	Indian Bengali	bn-Beng-IN	high_reach_underserved	FR-2	D2	SB-1	D3	MV-1 plus Introductory Statistics 2e	PM-S001
5	NAT-028	Vietnamese	vi-Latn-VN	high_reach_underserved	FR-2	D2	SB-1	D3	MV-1 plus Introductory Statistics 2e	ASSEC-S007
6	NAT-006	Marathi	mr-Dev-IN	high_reach_underserved	FR-2	D2	SB-1	D3	MV-1 plus Introductory Statistics 2e	PM-S001
7	NAT-004	Indian Tamil	ta-Tam-IN	high_reach_underserved	FR-2	D2	SB-1	D3	MV-1 plus Introductory Statistics 2e	PM-S001
8	NAT-015	Western Punjabi	pnb-Arab-PK	high_reach_underserved	FR-2	D2	MV-1	D2	Algebra and Trigonometry 2e	PM-S003
9	NAT-038	Javanese	jv-Latn-ID	high_reach_underserved	FR-2	D2	SB-1	D3	MV-1 plus Introductory Statistics 2e	PM-S005

Pos.	ID	Target	Profile	Lane	First comparator	First depth	Next comparator	Next depth	Next exact content	Population source
10	NAT-007	Gujarati	gu-Gujr-IN	high_reach_underserved	FR-2	D2	SB-1	D3	MV-1 plus Introductory Statistics 2e	PM-S001
11	NAT-029	Thai	th-Thai-TH	high_reach_underserved	FR-2	D2	SB-1	D3	MV-1 plus Introductory Statistics 2e	TH-S001
12	NAT-010	Odia	or-Orya-IN	high_reach_underserved	FR-2	D2	SB-1	D3	MV-1 plus Introductory Statistics 2e	PM-S001
13	NAT-008	Kannada	kn-Knda-IN	high_reach_underserved	FR-2	D2	SB-1	D3	MV-1 plus Introductory Statistics 2e	PM-S001
14	NAT-013	Indian Urdu	ur-Arab-IN	high_reach_underserved	FR-2	D2	SB-1	D3	MV-1 plus Introductory Statistics 2e	PM-S001
15	NAT-030	Burmese	my-Mymr-MM	high_reach_underserved	FR-2	D2	MV-1	D3	Algebra and Trigonometry 2e	ASSEC-S006
16	NAT-009	Malayalam	ml-Mlym-IN	high_reach_underserved	FR-2	D2	SB-1	D3	MV-1 plus Introductory Statistics 2e	PM-S001
17	EXP-003	Uzbekistan Uzbek Latin-script edition	uz-Latn-UZ	high_reach_underserved	FR-2	D2	SB-1	D3	MV-1 plus Introductory Statistics 2e	ALG-S018
18	NAT-014	Eastern Punjabi	pa-Guru-IN	high_reach_underserved	FR-2	D2	SB-1	D3	MV-1 plus Introductory Statistics 2e	PM-S001
19	NAT-039	Sundanese	su-Latn-ID	high_reach_underserved	FR-2	D2	SB-1	D3	MV-1 plus Introductory Statistics 2e	PM-S005
20	NAT-018	Sindhi	sd-Arab-PK	high_reach_underserved	FR-2	D2	MV-1	D2	Algebra and Trigonometry 2e	PM-S003
21	NAT-065	Amharic	am-Ethi-ET	high_reach_underserved	FR-2	D2	MV-1	D2	Algebra and Trigonometry 2e	PM-S020

Pos.	ID	Target	Profile	Lane	First comparator	First depth	Next comparator	Next depth	Next exact content	Population source
22	NAT-034	Cebuano	ceb-Latn-PH	high_reach_underserved	FR-2	D2	SB-1	D3	MV-1 plus Introductory Statistics 2e	ASSEC-S008
23	NAT-058	Dari	prs-Arab-AF	high_reach_underserved	FR-2	D2	MV-1	D2	Algebra and Trigonometry 2e	ASSEC-S018
24	NAT-012	Pakistani Urdu	ur-Arab-PK	high_reach_underserved	FR-2	D2	MV-1	D2	Algebra and Trigonometry 2e	PM-S003
25	NAT-011	Assamese	as-Beng-IN	high_reach_underserved	FR-2	D2	SB-1	D3	MV-1 plus Introductory Statistics 2e	PM-S001
26	NAT-019	Saraiki	skr-Arab-PK	high_reach_underserved	FR-2	D2	MV-1	D2	Algebra and Trigonometry 2e	PM-S003
27	NAT-027	Sinhala	si-Sinh-LK	high_reach_underserved	FR-2	D2	MV-1	D3	Algebra and Trigonometry 2e	ASSEC-S006
28	EXP-004	India Maithili Devanagari edition	mai-Deva-IN	high_reach_underserved	FR-2	D2	SB-1	D3	MV-1 plus Introductory Statistics 2e	PM-S001
29	EXP-001	Afghanistan Pashto Arabic-script edition	ps-Arab-AF	high_reach_underserved	FR-2	D2	MV-1	D2	Algebra and Trigonometry 2e	ASSEC-S018
30	NAT-022	Nepali	ne-Deva-NP	high_reach_underserved	FR-2	D2	MV-1	D2	Algebra and Trigonometry 2e	PM-S004
31	NAT-073	isiZulu	zu-Latn-ZA	high_reach_underserved	FR-2	D2	MV-1	D3	Algebra and Trigonometry 2e	ZA-S001
32	NAT-031	Khmer	km-Khmr-KH	high_reach_underserved	FR-2	D2	MV-1	D3	Algebra and Trigonometry 2e	ALG-S011
33	NAT-055	Turkmen	tk-Latn-TM	regional_depth	FR-2	D2	MV-1	D3	Algebra and Trigonometry 2e	ALG-S009

Pos.	ID	Target	Profile	Lane	First comparator	First depth	Next comparator	Next depth	Next exact content	Population source
34	NAT-075	Shona	sn-Latn-ZW	high_reach_underserved	FR-2	D2	MV-1	D2	Algebra and Trigonometry 2e	PM-S011
35	NAT-047	Kazakh	kk-Cyrl-KZ	high_reach_underserved	FR-2	D2	SB-1	D3	MV-1 plus Introductory Statistics 2e	ALG-S007
36	NAT-032	Lao	lo-Lao-LA	regional_depth	FR-2	D2	MV-1	D2	Algebra and Trigonometry 2e	ASSEC-S006
37	NAT-074	isiXhosa	xh-Latn-ZA	regional_depth	FR-2	D2	MV-1	D3	Algebra and Trigonometry 2e	ZA-S001
38	NAT-049	Tajik	tg-Cyrl-TJ	high_reach_underserved	FR-2	D2	SB-1	D3	MV-1 plus Introductory Statistics 2e	ASSEC-S006
39	EXP-013	Ethiopia Somali Latin-script edition	so-Latn-ET	regional_depth	FR-2	D2	MV-1	D2	Algebra and Trigonometry 2e	PM-S020
40	EXP-005	Indonesia Madurese Latin-script edition	mad-Latn-ID	regional_depth	FR-2	D2	MV-1	D3	Algebra and Trigonometry 2e	PM-S005
41	EXP-018	Mali Bamanankan Latin-script edition	bm-Latn-ML	regional_depth	FR-2	D2	MV-1	D2	Algebra and Trigonometry 2e	AFG-S008
42	NAT-048	Kyrgyz	ky-Cyrl-KG	regional_depth	FR-2	D2	MV-1	D3	Algebra and Trigonometry 2e	ALG-S008
43	NAT-035	Ilocano	ilo-Latn-PH	regional_depth	FR-2	D2	MV-1	D3	Algebra and Trigonometry 2e	ASSEC-S008
44	NAT-081	Wolof	wo-Latn-SN	regional_depth	FR-2	D2	MV-1	D2	Algebra and Trigonometry 2e	AFG-S009
45	NAT-005	Sri Lankan Tamil	ta-Taml-LK	regional_depth	FR-2	D2	MV-1	D3	Algebra and Trigonometry 2e	ASSEC-S006

Pos.	ID	Target	Profile	Lane	First comparator	First depth	Next comparator	Next depth	Next exact content	Population source
46	NAT-036	Hiligaynon	hil-Latn-PH	regional_depth	FR-2	D2	MV-1	D3	Algebra and Trigonometry 2e	ASSEC-S008
47	EXP-019	South Africa Sepedi Latin-script edition	nso-Latn-ZA	regional_depth	FR-2	D2	MV-1	D3	Algebra and Trigonometry 2e	ZA-S001
48	NAT-050	Mongolian	mn-Cyrl-MN	regional_depth	FR-2	D2	MV-1	D3	Algebra and Trigonometry 2e	ASSEC-S006
49	EXP-020	South Africa Setswana Latin-script edition	tn-Latn-ZA	regional_depth	FR-2	D2	MV-1	D3	Algebra and Trigonometry 2e	ZA-S001
50	EXP-021	South Africa Sesotho Latin-script edition	st-Latn-ZA	regional_depth	FR-2	D2	MV-1	D3	Algebra and Trigonometry 2e	ZA-S001
51	NAT-105	Maori	mi-Latn-NZ	regional_depth	FR-2	D2	MV-1	D3	Algebra and Trigonometry 2e	PM-S014
52	EXP-006	Indonesia Minangkabau Latin-script edition	min-Latn-ID	regional_depth	FR-2	D2	MV-1	D3	Algebra and Trigonometry 2e	PM-S005
53	EXP-014	Ethiopia Tigrinya Ethiopic-script edition	ti-Ethi-ET	regional_depth	FR-2	D2	MV-1	D2	Algebra and Trigonometry 2e	PM-S020
54	EXP-037	Ireland Irish Latin-script daily-use edition	ga-Latn-IE	small_population_prestige_domain_screen	FR-2	D2	MV-1	D3	Algebra and Trigonometry 2e	PM-S016
55	EXP-007	Indonesia Banjar Latin-script edition	bjn-Latn-ID	regional_depth	FR-2	D2	MV-1	D3	Algebra and Trigonometry 2e	PM-S005
56	OBS-POP-SN-AFG-002	Senegal Pulaar, official Latin orthography	fuc-Latn-SN	regional_depth	FR-2	D2	MV-1	D2	Algebra and Trigonometry 2e	AFG-S009
57	OBS-GRG-PL-007	Official-standard Belarusian	be-Cyrl-BY	small_population_prestige_domain_screen	FR-2	D2	MV-1	D3	Algebra and Trigonometry 2e	GRG-S006
58	NAT-037	Waray	war-Latn-PH	regional_depth	FR-2	D2	MV-1	D3	Algebra and Trigonometry 2e	ASSEC-S008

Pos.	ID	Target	Profile	Lane	First comparator	First depth	Next comparator	Next depth	Next exact content	Population source
59	EXP-015	Ethiopia Sidama Latin-script edition	sid-Latn-ET	regional_depth	FR-2	D2	MV-1	D2	Algebra and Trigonometry 2e	PM-S020
60	OBS-GRG-PL-010	Standard Lithuanian	lt-Latn-LT	small_population_prestige_domain_screen	FR-2	D2	MV-1	D3	Algebra and Trigonometry 2e	GRG-S006
61	EXP-008	Indonesia Sasak Latin-script edition	sas-Latn-ID	regional_depth	FR-2	D2	MV-1	D3	Algebra and Trigonometry 2e	PM-S005
62	EXP-022	South Africa Xitsonga Latin-script edition	ts-Latn-ZA	regional_depth	FR-2	D2	MV-1	D3	Algebra and Trigonometry 2e	ZA-S001
63	EXP-009	Indonesia Acehnese Latin-script edition	ace-Latn-ID	regional_depth	FR-2	D2	MV-1	D3	Algebra and Trigonometry 2e	PM-S005
64	NAT-021	Brahui	brh-Arab-PK	regional_depth	FR-2	D2	MV-1	D2	Algebra and Trigonometry 2e	PM-S003
65	NAT-079	Kirundi	rn-Latn-BI	regional_depth	FR-2	D2	MV-1	D2	Algebra and Trigonometry 2e	AFG-S011
66	EXP-010	Indonesia Betawi Latin-script edition	bew-Latn-ID	regional_depth	FR-2	D2	MV-1	D3	Algebra and Trigonometry 2e	PM-S005
67	NAT-023	Maithili Nepal profile	mai-Deva-NP	regional_depth	FR-2	D2	MV-1	D2	Algebra and Trigonometry 2e	PM-S004
68	EXP-023	South Africa siSwati Latin-script edition	ss-Latn-ZA	regional_depth	FR-2	D2	MV-1	D3	Algebra and Trigonometry 2e	ZA-S001
69	NAT-076	Zimbabwean Ndebele	nd-Latn-ZW	regional_depth	FR-2	D2	MV-1	D2	Algebra and Trigonometry 2e	PM-S011
70	EXP-016	Ethiopia Wolaytta Latin-script edition	wal-Latn-ET	regional_depth	FR-2	D2	MV-1	D2	Algebra and Trigonometry 2e	PM-S020
71	EXP-024	South Africa Tshivenda Latin-script edition	ve-Latn-ZA	regional_depth	FR-2	D2	MV-1	D3	Algebra and Trigonometry 2e	ZA-S001

Pos.	ID	Target	Profile	Lane	First comparator	First depth	Next comparator	Next depth	Next exact content	Population source
72	NAT-024	Bhojpuri Nepal profile	bho-Deva-NP	regional_depth	FR-2	D2	MV-1	D2	Algebra and Trigonometry 2e	PM-S004
73	EXP-029	Mauritius Mauritian Creole Latin-script edition	mfe-Latn-MU	regional_depth	FR-2	D2	MV-1	D3	Algebra and Trigonometry 2e	PM-S013
74	EXP-017	Ethiopia Hadiyya Latin-script edition	hdy-Latn-ET	regional_depth	FR-2	D2	MV-1	D2	Algebra and Trigonometry 2e	PM-S020
75	EXP-025	South Africa isiNdebele Latin-script edition	nr-Latn-ZA	regional_depth	FR-2	D2	MV-1	D3	Algebra and Trigonometry 2e	ZA-S001
76	OBS-GRG-US-025	Standard Haitian Creole	ht-Latn-HT	regional_depth	FR-2	D2	MV-1	D3	Algebra and Trigonometry 2e	GRG-S001
77	EXP-011	Indonesia Nias Latin-script edition	nia-Latn-ID	regional_depth	FR-2	D2	MV-1	D3	Algebra and Trigonometry 2e	PM-S005
78	NAT-026	Avadhi Nepal profile	awa-Deva-NP	regional_depth	FR-2	D2	MV-1	D2	Algebra and Trigonometry 2e	PM-S004
79	EXP-030	Zimbabwe Ndau Latin-script edition	ndc-Latn-ZW	regional_depth	FR-2	D2	MV-1	D2	Algebra and Trigonometry 2e	PM-S011
80	OBS-GRG-PL-002	Silesian in ślabikōrzowy szrajbōnek	szl-Latn-PL	regional_depth	FR-2	D2	MV-1	D3	Algebra and Trigonometry 2e	GRG-S006
81	NAT-041	Tetum Prasa	tet-Latn-TL	regional_depth	FR-2	D2	MV-1	D2	Algebra and Trigonometry 2e	ALG-S014
82	OBS-POP-SN-AFG-005	Senegal Mandinka (Màndienka), official Latin profile	mnk-Latn-SN	regional_depth	FR-2	D2	MV-1	D2	Algebra and Trigonometry 2e	AFG-S009
83	OBS-GRG-US-030	YIVO Standard Yiddish edition for the explicitly named YIVO-reading audience	ydd-Hebr-US	regional_depth	FR-2	D2	MV-1	D3	Algebra and Trigonometry 2e	GRG-S001

Pos.	ID	Target	Profile	Lane	First comparator	First depth	Next comparator	Next depth	Next exact content	Population source
84	OBS-GRG-US-029	English-oriented Pennsylvania German spelling used by Vitt Du Deitsh Shvetza / Di Heilich Shrift / Ich Kann PA Deitsh Shreiva	pdc-Latn-US	regional_depth	FR-2	D2	MV-1	D3	Algebra and Trigonometry 2e	GRG-S001
85	EXP-031	Timor-Leste Makasai Latin-script edition	mkz-Latn-TL	regional_depth	FR-2	D2	MV-1	D2	Algebra and Trigonometry 2e	ALG-S014
86	OBS-GRG-US-003	Diné Bizaad in Navajo Nation school orthography	nv-Latn-US	regional_depth	FR-2	D2	MV-1	D3	Algebra and Trigonometry 2e	GRG-S001
87	EXP-026	Peru Ashaninka Latin-script edition	cni-Latn-PE	small_population_prestige_domain_screen	FR-2	D2	MV-1	D2	Algebra and Trigonometry 2e	PM-S007
88	EXP-036	New Zealand Samoan Latin-script edition	sm-Latn-NZ	regional_depth	FR-2	D2	MV-1	D3	Algebra and Trigonometry 2e	PM-S014
89	EXP-027	Peru Awajun Latin-script edition	agr-Latn-PE	small_population_prestige_domain_screen	FR-2	D2	MV-1	D2	Algebra and Trigonometry 2e	PM-S007
90	OBS-GRG-PL-004	School-standard Kashubian	csb-Latn-PL	small_population_prestige_domain_screen	FR-2	D2	MV-1	D3	Algebra and Trigonometry 2e	GRG-S006
91	EXP-032	Timor-Leste Kemak Latin-script edition	kem-Latn-TL	small_population_prestige_domain_screen	FR-2	D2	MV-1	D2	Algebra and Trigonometry 2e	ALG-S014
92	OBS-GRG-US-026	Jamaican Creole in Cassidy/JLU orthography	jam-Latn-JM	small_population_prestige_domain_screen	FR-2	D2	MV-1	D3	Algebra and Trigonometry 2e	GRG-S001
93	EXP-033	Timor-Leste Bunak Latin-script edition	bfm-Latn-TL	small_population_prestige_domain_screen	FR-2	D2	MV-1	D2	Algebra and Trigonometry 2e	ALG-S014
94	EXP-034	Timor-Leste Tokodede Latin-script edition	tkd-Latn-TL	small_population_prestige_domain_screen	FR-2	D2	MV-1	D2	Algebra and Trigonometry 2e	ALG-S014
95	EXP-028	Peru Shipibo-Konibo Latin-script edition	shp-Latn-PE	small_population_prestige_domain_screen	FR-2	D2	MV-1	D2	Algebra and Trigonometry 2e	PM-S007

Pos.	ID	Target	Profile	Lane	First comparator	First depth	Next comparator	Next depth	Next exact content	Population source
96	NAT-044	Fataluku	ddg-Latn-TL	small_population_prestige_domain_screen	FR-2	D2	MV-1	D2	Algebra and Trigonometry 2e	ALG-S014
97	OBS-GRG-US-023	Standard 'Ōlelo Hawai'i with 'okina and kahakō	haw-Latn-US	small_population_prestige_domain_screen	FR-2	D2	MV-1	D3	Algebra and Trigonometry 2e	GRG-S001
98	OBS-GRG-US-017	Standard Marshallese under the 2010 Orthography Act	mh-Latn-MH	small_population_prestige_domain_screen	FR-2	D2	MV-1	D3	Algebra and Trigonometry 2e	GRG-S001
99	EXP-035	Timor-Leste Waima'a Latin-script edition	wmh-Latn-TL	small_population_prestige_domain_screen	FR-2	D2	MV-1	D2	Algebra and Trigonometry 2e	ALG-S014
100	OBS-GRG-US-001	Standard Central Alaskan Yup'ik	esu-Latn-US	small_population_prestige_domain_screen	FR-2	D2	MV-1	D3	Algebra and Trigonometry 2e	GRG-S001

F4. Needs-audited assignment layer

`top100_needs_assignment_v2.csv` replaces F3 for commissioning decisions. It contains a source-backed, territory- and stage-specific first-product or audit assignment for all 100 ranked rows, with confidence and caveats kept explicit. A row can recommend a bounded gap audit rather than a translation when current supply is already substantial or insufficiently inventoried. `population_mathematics_needs_register.csv` preserves the assessment population, indicator, source, curricular interpretation, confidence, and caveat behind the directly audited assignments.

Appendix G. Unresolved output profiles and D0 exclusions

ID	Issue	Target	Profile	Source population	Measure	Year	Population source	Why excluded	Evidence sought
EXP-002	D0_profile_population_mismatch	Pakistani Pashto	ps-Arab-PK	43633946	census_mother_tongue_persons	2023	PM-S003	exclude_D0_profile_population_mismatch; Pakistan census label Pushto is broader than an exact production variety	Exact production-variety/standard mapping within the PBS Pushto umbrella and a population-to-profile join that does not assign the whole category to one variety.
EXP-012	D0_profile_population_mismatch	Oromo	om-Latn-ET	24930424	census_mother_tongue_persons	2007	PM-S020	exclude_D0_profile_population_mismatch; census Oromigna category is broad and its editorial identifier is orm:CSA; exact variety reach is unresolved	Exact production-variety/standard mapping within the CSA Oromigna category, with cross-border populations excluded unless separately and disjointly evidenced.
OBS-GRG-US-004	unresolved_localized_output_profile	Eastern Keres	kee-Zzzz-US target profile; script unresolved by population source	12540	acs_2017_2021_estimated_persons_age5plus_language_spoken_at_home	2017-2021	GRG-S001	not rankable as one edition; split into community profiles first	Community-specific Eastern Keres variety, script, orthography, educational-standard authority, and a nonmultiplying population-to-output join.
OBS-GRG-US-012	unresolved_localized_output_profile	Tewa	tew-Zzzz-US target profile; script unresolved by population source	5105	acs_2017_2021_estimated_persons_age5plus_language_spoken_at_home	2017-2021	GRG-S001	not rankable as one edition; split into community profiles first	Pueblo-specific Tewa variety, script, orthography, educational-standard authority, and a nonmultiplying population-to-output join.
OBS-GRG-US-027	unresolved_localized_output_profile	Guyanese Creole English	gyn-Latn-GY	4603	acs_2017_2021_estimated_persons_age5plus_language_spoken_at_home	2017-2021	GRG-S001	rank shared core and localized profiles separately; do not multiply one population observation	Exact institutional teaching orthography and an audience split between a shared core and localized profiles without multiplying GRG-US-027.

These are exclusions from recommendation ranking, not claims that the communities lack educational need. The two D0 rows retain source population cells but do not assign them to an exact production profile.

Appendix H. Interlanguage overlap matrix summary

The complete registered matrix contains 104 rows: 80 interlanguage rows across 15 intervention IDs and 24 accessibility rows across 18 IDs. IL-AR occurs in both mechanism scopes, so those ID counts are nonadditive and reconcile to 32 distinct IDs overall. Appendix F summarizes only the 80 interlanguage rows; Appendix C handles accessibility. Every interlanguage component remains non-rankable under current evidence, and every demographic or component subtotal is nonadditive unless an explicit disjoint-universe rule says otherwise.

ID	Mechanism	Row s	Exact communities/varietie s	Scripts	Task evidence	Prior-study evidence	Existing-edition overlap	Exclusions	Double-count rule
IL-AR	pluricentric_s hared_core	1	Readers comfortable with Modern Standard Arabic	Arab	none registered for target corpus	formal literacy required	complete local Arabic edition	Amazigh Kurdish Nubian Modern South Arabian and signed languages	Existing MSA coverage is baseline and cannot be claimed again by scaffolds
IL- BCMS	dual_script_r egister pluricentric_s hared_core	5	Bosnian-standard cell Croatian-standard cell Montenegrin- standard cell Serbian Cyrillic- reading stratum Serbian Latin-reading stratum	Cyrl Latn	none registered	none beyond named-standard literacy none beyond script literacy	none established	Croatian Montenegrin Serbian and non- BCMS languages remain distinct cells Other BCMS standards Other BCMS standards remain distinct cells	Named output claims only Bosnian cell; shared source has compute value only Named output claims only Croatian cell Named output claims only Montenegrin cell Partition one Serbian population by script use; do not add Latin and Cyrillic

ID	Mechanism	Row s	Exact communities/varietie s	Scripts	Task evidence	Prior-study evidence	Existing-edition overlap	Exclusions	Double-count rule
IL- GER M	natural_inter comprehensi on_reuse	10	Afrikaans-standard cell Dutch- standard cell Faroese cell Hutterisch cell Low German exact- standard cell Pennsylvania Dutch cell Plautdietsch cell Scots cell West Frisian cell Yiddish exact-register cell	Hebr Latn	none	none beyond community- standard literacy none beyond exact-register literacy none beyond exact- standard literacy none beyond named-standard literacy	Danish and English overlap unmeasured Dutch and English overlap unmeasured English and German overlap unmeasured English Hebrew Russian and German overlaps vary English overlap expected high but unmeasured English overlap expected very high but unmeasured English overlap likely high but unmeasured German English Russian Spanish and Portuguese overlaps vary German overlap unmeasured	All Latin-script Germanic profiles All residual Germanic communities Dutch Afrikaans and other Germanic communities Mainland Scandinavian and other Germanic languages Other Frisian languages and Germanic communities Other Germanic communities Other Germanic diaspora languages Other Germanic languages	Claims only residual Afrikaans cell; directional Dutch credit zero Claims only residual Dutch cell; cross-language credit zero Count only residual comfortable-reader gain; no pan- Germanic credit Count only residual community readers; no pan-Germanic credit Deduplicate by residence and dominant academic- language access; no pan-Germanic credit No pan-Germanic credit No pan- Germanic credit; named output only No pan-Germanic credit; residence and community register define separate cells No pan-Germanic or mainland Scandinavian credit

ID	Mechanism	Row s	Exact communities/varietie s	Scripts	Task evidence	Prior-study evidence	Existing-edition overlap	Exclusions	Double-count rule
IL-HU	dual_script_ register	3	Indian Urdu cell Pakistani Urdu cell Standard Hindi- oriented cell	Arab Nastaliq Deva	none	none beyond named-standard literacy	C-17 reports English- or-Hindi subsidiary- language knowledge for 23172258 of 50772631 Urdu- category speakers; academic comfort remains unknown complete local Hindi edition; C-17 reports English subsidiary- language knowledge for 35257144 of the 528347193 Hindi census umbrella Hindi overlap unmeasured	Other Indian mother tongues Punjabi Sindhi Saraiki Pashto and other languages Urdu and Hindi census umbrella sublanguages	Existing Hindi coverage is baseline and not added to Urdu adaptation reach; C-17 language knowledge is not academic comfort Hindi gives zero Urdu credit; named Urdu output claims exact Pakistan cell Keep Indian Urdu separate from Pakistan Urdu; apply disclosed nonoverlap sensitivity 0.543607303/0.77180 3651/0.908721461 rather than categorical Hindi coverage
IL- IDMS	dual_script_ register pluricentric_ shared_core	4	Brunei Malay educational-standard cell Indonesian standard cell Malaysian Malay Jawi-reading stratum Malaysian Malay Rumi cell	Arab Latn	none	none beyond exact-standard literacy none beyond Jawi literacy none beyond named-standard literacy	722 of 722 Open Logic units complete Indonesian and Malaysian overlap untested Indonesian cross- standard overlap untested Same language as ms-Latn- MY	Javanese Sundanese Madurese Minangkabau Acehnese and other languages Other Malayic and Indonesian languages Other Malayic languages	Completed Indonesian Open Logic is excluded from forward reach; no Malaysian reach Named Brunei cell only; no regional blanket credit Partition or model dual literacy; never add Jawi and Rumi populations Zero Indonesian credit pending directional technical-text audit; localized output claims Malaysian cell

ID	Mechanism	Row s	Exact communities/varietie s	Scripts	Task evidence	Prior-study evidence	Existing-edition overlap	Exclusions	Double-count rule
IL-ISV	constructed_ bridge	15	Belarusian-literate cohort Bosnian-standard cohort Bulgarian-literate cohort Croatian-standard cohort Czech-literate cohort Macedonian-literate cohort Montenegrin-standard cohort Polish-literate cohort Pooled native-Slavic study respondents Russian-literate cohort Serbian Cyrillic-script reading cell Serbian Latin-script reading cell Slovak-literate cohort Slovene-literate cohort Ukrainian-literate cohort	Cyrl Latn study surface reported as written Interslavic	none at exact leaf level in current register seven-gap written professional-text cloze [seven-gap short written professional-text cloze] score=0.84	not required for the short cloze; sustained mathematics unknown unknown for sustained mathematics	No exact local native-language-edition overlap was subtracted unknown unknown in current local coverage census	Kashubian is not included Kashubian Sorbian Rusyn and all untested minority leaves No additive credit with Serbian Cyrillic No additive credit with Serbian Latin No BCMS or other South-Slavic transfer credit No Bulgarian or other South-Slavic transfer credit No Czech or other West-Slavic transfer credit No Macedonian or other South-Slavic transfer credit No Slovak or other West-Slavic transfer credit No unnamed BCMS population aggregation No untested minority or exact language leaf receives this pooled value Untested Slavic leaves	Credit zero until an exact surface/script/cohort test; deduct any comfortable Russian edition before reach Credit zero until exact testing and deduct any comfortable Ukrainian edition Partition one Serbian population by observed script use; then deduct named Serbian access The pooled 0.84 is one study-level point estimate and may not be copied to leaf rows or multiplied by a Slavic demographic ceiling Zero until exact testing and after any named BCMS edition Zero until exact testing and overlap subtraction Zero until exact testing; deduct Czech and English comfortable access Zero until exact testing; deduct Polish and English comfortable access Zero until exact testing; deduct Slovak and English comfortable access Zero until exact testing; deduct Slovene and English comfortable access Zero until exact testing; one person cannot be counted in both Latin and Cyrillic projection cells

ID	Mechanism	Row s	Exact communities/varietie s	Scripts	Task evidence	Prior-study evidence	Existing-edition overlap	Exclusions	Double-count rule
IL- MAN DING	dual_script_r egister pluricentric_s hared_core	5	Bambara Latin- standard cell Bambara N'Ko-literate stratum Julia Latin- standard cell Maninka exact- national-standard cell Maninka N'Ko- literate stratum	Latn Nkoo	none registered	none beyond exact-standard literacy none beyond N'Ko literacy none beyond named-standard literacy	none established Same Bambara language cell as Latin output Same Maninka language cell as Latin output	Other Manding and Mande languages Other Manding languages	Named output and country cell only Named output and territory cell only Named output only; no continuum-wide population Partition by observed script use; never add Latin and N'Ko users
IL- NGU NI	pluricentric_s hared_core	4	isiXhosa-standard cell isiZulu-standard cell siSwati- standard cell Southern Ndebele standard cell	Latn	none registered	none beyond named-standard literacy	none established	isiXhosa siSwati isiNdebele and non- Nguni languages Other Nguni standards Zimbabwean Ndebele and other Nguni standards	Country cells and named output only Keep nr and nd distinct; named output only Named output only; no combined Nguni population Named output only; no cross- language credit
IL- PDT	pluricentric_s hared_core	3	Dari-standard cell Iranian Persian cell Tajik-standard cell	Arab Cyril	none	none beyond named-standard literacy	complete local Iranian Persian edition Iranian Persian directional overlap unmeasured	Dari Tajik Kurdish Pashto Balochi and Pamiri languages Pamiri Uzbek Russian and other languages Pashto Hazaragi and other languages	Existing fa-IR coverage is baseline and not added to adaptation reach Zero fa-IR reach pending technical- text evidence; named Dari output claims only exact cells Zero fa-IR reach; named Tajik output only; transliteration is not additive
IL- PUNJ ABI	dual_script_r egister	2	Eastern Punjabi Gurmukhi cell Western Punjabi Shahmukhi cell	Arab Shahmukhi Guru	none	none beyond named-standard literacy	C-17 reports English- or-Hindi subsidiary- language knowledge for 17571723 of 33124726 Punjabi- category speakers; academic comfort remains unknown none established	Saraiki Hindko and unresolved Punjabi census sublabels Western Punjabi Saraiki Hindko	Claims only India Gurmukhi cell; apply nonoverlap sensitivity 0.469528503/0.73476 4251/0.893905701 and never add a dual- script population Claims only resolved Shahmukhi population; no additive dual-script population

ID	Mechanism	Rows	Exact communities/varieties	Scripts	Task evidence	Prior-study evidence	Existing-edition overlap	Exclusions	Double-count rule
IL-ROM	constructed_bridge pluricentric_shared_core	8	Brazilian Portuguese cell Catalan-standard cell French-standard cell Galician-standard cell Italian-standard cell Proposed constructed pan-Romance surface Romanian-standard cell Spanish-standard cell	Latn	none none required for own localized output	none beyond literacy in exact locale standard none beyond literacy in exact standard none beyond literacy in named standard none beyond literacy in the named standard unknown	complete local edition not established not established in current successor census Spanish and Brazilian Portuguese complete locally	All named Romance languages absent exact testing Aromanian and other Eastern Romance varieties Catalan Galician and other languages are not Spanish cells European and African Portuguese profiles not automatically included Occitan and other Romance languages Other Romance languages Portuguese and Spanish are separate baseline editions Sardinian Sicilian Neapolitan Friulian and Ladin	Claims only Catalan cell; Spanish overlap deducted by cohort Claims only exact French cell; bilingual overlap deducted separately Claims only exact Romanian cell Claims only Italian cell Claims only residual Galician cell after measured Spanish/Portuguese comfort Constructed surface has zero coverage; design artifacts cannot claim demographic ceilings pt-BR output is subtracted from residual bridge scenarios; no additive bridge credit Spanish output claims only exact Spanish residual cells and is subtracted before any bridge scenario
IL-SCAND	natural_intercomprehension_reuse	4	Danish-standard cell Norwegian Bokmal cell Norwegian Nynorsk cell Swedish-standard cell	Latn	exact pair values not registered	none beyond named-standard literacy	Bokmal English and Scandinavian overlap unmeasured English and Scandinavian overlap unmeasured	Faroese Icelandic Nynorsk Faroese Icelandic	Named output only; cross-language reach zero until exact directional value is imported Named output only; no cross-language credit Partition Norwegian readers and do not add dual-standard literacy
IL-SOTHO	pluricentric_shared_core	3	Northern Sotho standard cell Southern Sotho standard cell Tswana standard cell	Latn	none registered	none beyond named-standard literacy	none established	Other Sotho-Tswana standards Tswana Southern Sotho Venda Tsonga and Nguni languages	Country cells remain separate; named output only Named output only; no combined Sotho-Tswana population

ID	Mechanism	Row s	Exact communities/varietie s	Scripts	Task evidence	Prior-study evidence	Existing-edition overlap	Exclusions	Double-count rule
IL- SQUE CHUA	pluricentric_ shared_core	3	Ayacucho Quechua cell Cusco Quechua cell South Bolivian Quechua cell	Latn	none registered	none beyond named-variety literacy	Spanish overlap unmeasured	Ayacucho South Bolivian Northern and Central Quechua Cusco South Bolivian Northern and Central Quechua Peruvian Southern Northern and Central Quechua	Named variety and country cell only; no aggregate Quechua population Named variety only; no aggregate Quechua population
IL- TURK IC	natural_inter comprehensi on_reuse	10	Azerbaijani-standard cell Bashkir- standard cell Gagauz-standard cell Kazakh-standard cell Kyrgyz- standard cell Tatar- standard cell Turkish-standard cell Turkmen-standard cell Uyghur- standard cell Uzbek-standard cell	Arab Cyr Latn	none	none beyond named-standard literacy	complete local Turkish edition Russian and Turkic overlap unmeasured Turkish overlap unmeasured Uzbek and Turkic overlap unmeasured	All other Turkic languages Oghuz and other Kipchak languages Other Karluk profiles Other Kipchak languages Other Oghuz languages Uyghur and all other Karluk profiles	Cross-language reach zero; named output only Cross- language Turkish credit is zero; named Azerbaijani output claims only Azerbaijani cell No cross-language credit No cross-language credit; legacy Cyrillic is a script-access cell No cross-language credit; script projection is not additive Turkish population is already covered locally and is never added to branch reuse reach

The Interslavic short-cloze evidence remains a task-specific observed result; it is not a sustained mathematics-comprehension estimate and is never multiplied by a Slavic-family demographic total.

H2. Top-100 overlap-control result

The separate `top100_interlanguage_overlap_crosswalk.csv` joins all 100 ordered natural-language interventions to this matrix. It records 16 exact-profile matches, four exact language/script plus named-country matches, one hypothesis-only relation, and 79 unmapped rows. Every cross-language demographic reach credit is zero, every interlanguage-rankability flag is false, and positive shared-core reuse remains unquantified. These zeros are overlap-control values, not estimates of mathematical utility. The crosswalk's SHA-256 is `ED88A556E1055655E3635AAA64C793A684E1C655713BA4F86AD50B851CC1A380`.

Appendix I. Table-source authorities and source notes

Generated from the exact source IDs used by the current Table 2 and Top 100 snapshots. Draft references use only locally registered metadata; each entry retains its authorship basis and bounded source note.

Population and count authorities

AFG-S008

Institut National de la Statistique (INSTAT) and Bureau Central du Recensement (BCR), Mali. (2022). *RGPH5 2022: Caractéristiques culturelles de la population* [official census thematic report]. <https://bibliostat.instat.ml/greenstone3/library/collection/dmograph/document/HASH01e98829c5a7e2847f521a98?ed=1>

Authorship basis: registered authority used as corporate author

Source note: AFG-S008 supplies Resident population aged 3 years and above by mother-tongue category for Mali. Registered locator: Tableau 3.01, printed p.25; HTML document node HASH01e98829c5a7e2847f521a98.40. Registered temporal field: 2022. Caveat: Table 3.01 and Annex A03 publish materially different counts, and some printed sex subtotals do not exactly add to the printed ensemble cells. These observations preserve the Table 3.01 ensemble cells verbatim and do not repair or combine them. Peulh/Fulfulde and Malinké/Maninkakan remain source aggregates unless separately resolved.

AFG-S009

Agence Nationale de la Statistique et de la Démographie (ANSD), Senegal. (2023). *RGPH-5 2023 definitive report, Theme I: Etat et structure, urbanisation et caractéristiques socioculturelles de la population* [official census report]. https://www.ansd.sn/sites/default/files/recensements/rapport/rapport_national/Rapport-def-RGPH-5.pdf

Authorship basis: registered authority used as corporate author

Source note: AFG-S009 supplies Resident population aged 3 years and above by principal language commonly spoken for Senegal. Registered locator: Tableau I-32, printed pp.57-58; PDF pp.94-95. Registered temporal field: 2023. Caveat: Principal-language categories are mutually exclusive and do not

measure all languages spoken, literacy, or academic reading ability. Joola is not a single exact variety. ISO mappings are editorial and not supplied by the census table.

AFG-S011

Institut de Statistiques et d'Études Économiques du Burundi (ISTEEBU/INSBU). (2008). *Répartition de la population issue du RGPH 2008* [official census tables]. <https://www.insbu.bi/documents/recensements/R%C3%A9partition-de-la-population-issue-du-RGPH-2008.pdf>

Authorship basis: registered authority used as corporate author

Source note: AFG-S011 supplies Population aged 10 years and above by mutually exclusive language read-and-written combination for Burundi. Registered locator: Tableau 3.2, PDF p.62. Registered temporal field: 2008. Caveat: The any-Kirundi count is a derived sum of 16 mutually exclusive total-column rows containing Kirundi. It is nested over the exact Kirundi-only cell; never add the two observations. Counts concern reading and writing among persons aged 10+, not total speakers.

ALG-S007

Bureau of National Statistics of the Agency for Strategic Planning and Reforms of the Republic of Kazakhstan. (2021). *National composition, religion and language proficiency in the Republic of Kazakhstan* [official national census statistical collection]. <https://stat.gov.kz/upload/medialibrary/cee/3rsfg8ps3xo19orb284esg4rx27ihqf7/%D0%9D%D0%B0%D1%86%D0%B8%D0%BE%D0%BD%D0%B0%BB%D1%8C%D0%BD%D1%8B%D0%B9%20%D1%81%D0%BE%D1%81%D1%82%D0%B0%D0%B2.pdf>

Authorship basis: registered authority used as corporate author

Source note: ALG-S007 supplies Native language by nationality and population language proficiency for Kazakhstan. Registered locator: Table 6, printed p. 243; Table 8, printed p. 366. Registered temporal field: 2021. Caveat: Table 6 counts language-of-own-nationality only within each nationality and omits cross-nationality native speakers. Table 8 proficiency is not mother tongue or academic comfort; its 17,194,712-person universe appears to be age 5+ from the age table, so the age interpretation is retained as a caveat.

ALG-S008

National Statistical Committee of the Kyrgyz Republic. (2022). *Population and Housing Census of the Kyrgyz Republic 2022: Book II (Part One), Population of Kyrgyzstan* [official national census statistical collection]. <https://stat.gov.kg/media/publicationarchive/0f717d2d-5078-4e18-8b56-fb2a530462a2.pdf>

Authorship basis: registered authority used as corporate author

Source note: ALG-S008 supplies Native language by ethnic group for Kyrgyzstan. Registered locator: Table 3.8, printed pp. 118-119. Registered temporal field: 2022. Caveat: National Kyrgyz, Uzbek, and Russian native-language totals in the observation file are exact sums of mutually exclusive own-ethnicity and cross-ethnicity cells. Other named groups are nationality-scoped own-language counts and therefore lower bounds on total speakers.

ALG-S009

State Committee of Turkmenistan on Statistics. (2022). *Results of the Complete Population and Housing Census of Turkmenistan - 2022: National Composition and Language Proficiency* [official national census tables]. <https://stat.gov.tm/population-census-pdfs/results/en/4.pdf>

Authorship basis: registered authority used as corporate author

Source note: ALG-S009 supplies Population by mother tongue for Turkmenistan. Registered locator: Table 4.9, PDF section pp. 8-10. Registered temporal field: 2022. Caveat: Direct mother-tongue totals. Published category labels such as Baloch are retained as-is and do not establish a narrower variety.

ALG-S011

National Institute of Statistics Cambodia. (2019). *General Population Census of the Kingdom of Cambodia 2019: National Report on Final Census Results* [official national census report]. <https://nis.gov.kh/nis/Census2019/Final%20General%20Population%20Census%202019-English.pdf>

Authorship basis: registered authority used as corporate author

Source note: ALG-S011 supplies Population by mother tongue for Cambodia. Registered locator: Table 2.7.1, PDF p. 53 / printed p. 25. Registered temporal field: 2019. Caveat: The table covers all persons in census households and excludes migrants working abroad. Published total-category counts sum to the stated universe, but multiple male/female subtotals do not sum to their published category totals; only the total-category column is staged.

ALG-S014

General Directorate of Statistics Timor-Leste. (2015). *Timor-Leste Population and Housing Census 2015 Volume 2 Language Tables* [official national census workbook]. <https://www.laohamutuk.org/DVD/DGS/Cens15/Census2015v2data.zip>

Authorship basis: registered authority used as corporate author

Source note: ALG-S014 supplies Population by mother tongue for Timor-Leste. Registered locator: Workbook 4_2015 V2 Language.xls, sheet 2.12, Table 12. Registered temporal field: 2015. Caveat: The official DGS workbook is preserved from a longstanding external mirror archive. Counts are direct mother-tongue totals; distinct source varieties such as Tetun Prasa and Tetun Terik remain separate.

ALG-S018

National Statistics Committee of the Republic of Uzbekistan. (2026, June 30). *Проведена конференция посвящённая предварительным результатам переписи населения и сельского хозяйства* [official preliminary national census release]. <https://stat.uz/ru/press-tsentr/novosti-goskomstata/68978-a-oliva-ishlo-kh-zhaligini-r-jkhatga-olish-tadbirining-dastlabki-natizhalariga-ba-ishlangan-konferentsiya-tkazildi-3>

Authorship basis: registered authority used as corporate author

Source note: ALG-S018 supplies Preliminary national population and Uzbek mother-tongue results for Uzbekistan. Registered locator: Paragraphs reporting the exact preliminary population total and 35.7 million / 91.3 percent Uzbek mother tongue. Registered temporal field: 2026-06-30. Caveat: Preliminary release. The mother-tongue count is rounded to 0.1 million; its separately published 91.3 percent does not arithmetically reproduce 35.7 million against the exact preliminary population total, so the percentage is not used to narrow the count range. Post-enumeration coverage was reported as 97.3 percent.

ASSEC-S006

Transparent CLDR 48.2 target person-count derivations and exclusion audit. (2026, March 17). [transparent secondary derivation witness]. <https://raw.githubusercontent.com/unicode-org/clldr/release-48-2/common/supplemental/supplementalData.xml>

Authorship basis: title-first reference because the register identifies an arithmetic derivation, not an author

Source note: ASSEC-S006 supplies Nearest-person arithmetic from pinned territory population times pinned language population percentage; no invented confidence interval for Bangladesh; Myanmar; Laos; Mongolia; Sri Lanka; Malaysia; Tajikistan. Registered locator: cldr_target_derivations.md formulas table and exclusion audit. Registered temporal field: 2026-03-17 release; component years not stated. Caveat: Derived point counts inherit CLDR's unspecified component vintages and uncertainty. Integer percentage rounding rules are not published, so observation low/high fields remain blank. Not direct census evidence.

ASSEC-S007

Phạm, B., & McLeod, S. (2016). Consonants, vowels and tones across Vietnamese dialects. *International Journal of Speech-Language Pathology*, 18(2), 122-134.
<https://doi.org/10.3109/17549507.2015.1101162>

Authorship basis: named authors, journal, volume, issue, pages, and DOI read from the hash-verified local PubMed XML witness

Source note: ASSEC-S007 supplies Published lower-bound statement that Vietnamese is spoken by over 89 million people in Vietnam for Vietnam. Registered locator: PMID 27172848; AbstractText label PURPOSE first sentence; volume 18 issue 2 pages 122-134; DOI 10.3109/17549507.2015.1101162. Registered temporal field: 2016; estimate vintage not stated. Caveat: Rounded lower bound rather than a point estimate; L1 versus L2 and source year of the population estimate are not stated. The article is about dialect phonetics rather than population estimation.

ASSEC-S008

CLEAR Global. (2010). *CLEAR Global Language Use Data Platform Philippines national API snapshot* [Weighted census-microdata API]. https://ludp.clearglobal.org/public/location/PHL/?aggregation=0&fields=proportion_value,language_rank,language_name,language_code,location_name,location_code,location_level,dataset_name,url,source,datetime_published,reliability_score,individuals_value_weighted&page_size=500

Authorship basis: CLEAR Global selected from the registered compound authority as the API platform/corporate author; underlying IPUMS Census 2010 provenance remains in the source note

Source note: ASSEC-S008 supplies Weighted persons by normalized main language at national aggregation level zero for Philippines. Registered locator: JSON data rows Cebuano cebu1242; Iloko ilok1237; Hiligaynon hili1240; Waray (Philippines) wara1300; fields individuals_value_weighted and reliability_score. Registered temporal field: 2010-12-31. Caveat: Exact API point values are weighted estimates, not exact headcounts; no variance or confidence interval is exposed. The platform normalizes the metric as main language. API content is mutable, so the local byte witness is controlling.

ASSEC-S018

Transparent Afghanistan Dari and Pashto person-count derivation. (2020). [transparent secondary derivation witness]. https://web.archive.org/web/20230102023706id_/https://www.cia.gov/the-world-factbook/countries/afghanistan/

Authorship basis: title-first reference because the register identifies an arithmetic derivation, not an author

Source note: ASSEC-S018 supplies Nearest-person product of World Bank 2020 population and CIA 2020 total-speaker shares, with official report semantic check for Afghanistan. Registered locator: afghanistan_language_derivation.md formulas for Dari and Pashto. Registered temporal field: 2020. Caveat: Derived point counts inherit the source shares' unknown rounding and uncertainty. No numerical low/high interval is invented. Dari and Pashto totals overlap and must never be summed; displaced cohorts outside Afghanistan are excluded.

GRG-S001

U.S. Census Bureau. (n.d.). *Detailed Languages Spoken at Home and Ability to Speak English for the Population 5 Years and Over in the United States: 2017-2021* [official ACS five-year detailed-language tabulation]. Retrieved August 25, 2026, from <https://www2.census.gov/programs-surveys/demo/tables/language-use/2017/2017-2021-acslang-tables-nation.xlsx>

Authorship basis: registered authority used as corporate author

Source note: GRG-S001 supplies Estimated persons age 5 years and over speaking the named language at home with published 90 percent margin of error for United States. Registered locator: Nation worksheet; selected indented child rows 11, 22, 24, 231, 233-234, 237-239, 246-247, 255, 260, 272, 458, 462, 473-474, 487-488, 492, 508, 510-511, 520, 522, 527, 529, 535, 538, 547-548, 552, 554, 564; notes rows 575-581. Registered temporal field: 2017-2021. Caveat: Sample estimates, not enumerated counts. Universe is age 5 and over and the measure is language spoken at home, not mother tongue or educational access. Parent language-family rows overlap child rows and were excluded. Published margins of error are 90 percent intervals; nonsampling error is not included. Detailed estimates may not sum to aggregates because of rounding.

GRG-S002

Australian Bureau of Statistics. (2021). *2021 Lingiari, Census All persons QuickStats* [official national census QuickStats]. <https://www.abs.gov.au/census/find-census-data/quickstats/2021/CED701>

Authorship basis: registered authority used as corporate author

Source note: GRG-S002 supplies Persons reporting the named main language used at home in the national Australia comparison column for Australia. Registered locator: Language used at home, top responses other than English; Australia column; Kriol, Djambarrpuyngu, Warlpiri, Murrinh Patha and

Alyawarr rows. Registered temporal field: 2021. Caveat: National values are the Australia comparison column, not the Lingiari local counts. ABS applied small random changes to cells for privacy. Language used at home permits one main-language response and is not an L1 or educational-access measure.

GRG-S003

Australian Bureau of Statistics. (2021). *2021 Alice Springs, Census All persons QuickStats* [official national census QuickStats]. <https://www.abs.gov.au/census/find-census-data/quickstats/2021/70201>

Authorship basis: registered authority used as corporate author

Source note: GRG-S003 supplies Persons reporting the named main language used at home in the national Australia comparison column for Australia. Registered locator: Language used at home, top responses other than English; Australia column; Pitjantjatjara row. Registered temporal field: 2021. Caveat: National value is the Australia comparison column, not the Alice Springs local count. ABS applied small random changes to cells for privacy. Language used at home permits one main-language response and is not an L1 or educational-access measure.

GRG-S004

Australian Bureau of Statistics. (2021). *2021 Katherine, Census All persons QuickStats* [official national census QuickStats]. <https://www.abs.gov.au/census/find-census-data/quickstats/2021/70205>

Authorship basis: registered authority used as corporate author

Source note: GRG-S004 supplies Persons reporting the named main language used at home in the national Australia comparison column for Australia. Registered locator: Language used at home, top responses other than English; Australia column; Gurindji, Ngarinyman and Nunggubuyu rows. Registered temporal field: 2021. Caveat: National values are the Australia comparison column, not the Katherine local counts. ABS applied small random changes to cells for privacy. Language used at home permits one main-language response and is not an L1 or educational-access measure.

GRG-S005

Australian Bureau of Statistics. (2021). *2021 Barkly, Census All persons QuickStats* [official national census QuickStats]. <https://www.abs.gov.au/census/find-census-data/quickstats/2021/70202>

Authorship basis: registered authority used as corporate author

Source note: GRG-S005 supplies Persons reporting the named main language used at home in the national Australia comparison column for Australia. Registered locator: Language used at home, top responses other than English; Australia column; Warumungu row. Registered temporal field: 2021.

Caveat: National value is the Australia comparison column, not the Barkly local count. ABS applied small random changes to cells for privacy. Language used at home permits one main-language response and is not an L1 or educational-access measure.

GRG-S006

Statistics Poland. (2021, March 31). *Size and demographic-social structure in the light of the 2021 Census results* [official national census publication].
https://stat.gov.pl/files/gfx/portalinformacyjny/en/defaultaktualnosci/3701/6/1/1/national_population_and_housing_census_2021_population_2.pdf

Authorship basis: registered authority used as corporate author

Source note: GRG-S006 supplies Persons using the named language at home; respondents could indicate more than one language for Poland. Registered locator: Table 7.5, printed p. 120, PDF p. 122. Registered temporal field: 2021-03-31. Caveat: Multiple language responses were allowed, so rows overlap and must not be summed. The table measures language used at home, not L1 or academic-language comfort. Official bilingual labels are retained exactly. The population source does not report scripts or orthographies.

PM-S001

Office of the Registrar General & Census Commissioner, India. (2011). *C-16 Population by Mother Tongue, Statement 1: Abstract of Speakers' Strength of Languages and Mother Tongues, Census of India 2011* [official census table].
https://censusindia.gov.in/nada/index.php/catalog/42458/download/46089/C-16_25062018.pdf

Authorship basis: registered authority used as corporate author

Source note: PM-S001 supplies Persons returned under each census language or mother-tongue category for India. Registered locator: Statement 1, printed pp. 5-12; Part A scheduled-language totals and named mother-tongue components. Registered temporal field: 2011. Caveat: Census language categories can aggregate distinct mother tongues; Hindi parent total overlaps its listed child categories and must not be summed with them. Counts are 2011 observations, not current estimates.

PM-S003

Pakistan Bureau of Statistics. (2023a). *Table 11: Population by Mother Tongue, Sex and Rural/Urban, Census 2023* [official census table].
https://www.pbs.gov.pk/wp-content/uploads/census_tables/tables/table_11_national.pdf

Authorship basis: registered authority used as corporate author

Source note: PM-S003 supplies All-localities all-sexes persons by reported mother tongue for Pakistan. Registered locator: Table 11, national row, PDF p. 1. Registered temporal field: 2023. Caveat: Several labels are census aggregates (for example Pushto, Balochi, Kohistani); they are not automatically single edition targets.

PM-S004

National Statistics Office, Nepal. (2021). *Languages in Nepal: National Population and Housing Census 2021 Thematic Report* [official census thematic report].
<https://censusnepal.cbs.gov.np/results/files/result-folder/Language%20in%20Nepal.pdf>

Authorship basis: registered authority used as corporate author

Source note: PM-S004 supplies Persons by reported mother tongue for Nepal. Registered locator: Annex 1, printed pp. 123-126. Registered temporal field: 2021. Caveat: Some census labels aggregate varieties; the Sign Language entry is a census mother-tongue response category rather than a full sign-language-use census.

PM-S005

BPS-Statistics Indonesia. (2010). *Kewarganegaraan, Suku Bangsa, Agama, dan Bahasa Sehari-hari Penduduk Indonesia: Hasil Sensus Penduduk 2010* [official census publication].
<https://www.bps.go.id/en/publication/2012/05/23/55eca38b7fe0830834605b35/nationality--ethnicity--religion--and-dailylanguage-of-indonesian-population.html>

Authorship basis: registered authority used as corporate author

Source note: PM-S005 supplies Population aged 5 and over by language used daily at home for Indonesia. Registered locator: Table L4.1, printed p. 47 (PDF p. 55). Registered temporal field: 2010. Caveat: Home-use counts are not L1 totals. Several rows are broad language groups, including Batak and regional clusters; only individually named cells are retained here.

Local evidence status: No local witness path or hash is registered; do not describe this source as locally hash-verified.

PM-S007

Instituto Nacional de Estadística e Informática (INEI). (2017). *Peru: Perfil Sociodemográfico, Censos Nacionales 2017* [official census report].
https://www.inei.gob.pe/media/MenuRecursivo/publicaciones_digitales/Est/Lib1544/00TOMO_01.pdf

Authorship basis: registered authority used as corporate author

Source note: PM-S007 supplies Census population aged 3 and over by language or mother tongue learned in childhood for Peru. Registered locator: Cuadro 4, printed p. 618, total column. Registered temporal field: 2017. Caveat: Quechua and Aimara are aggregate census labels spanning varieties. The total column is used from a health-insurance cross-tabulation; it remains the national count for the stated age and language universe.

PM-S011

Zimbabwe National Statistics Agency (ZIMSTAT). (2022). *Zimbabwe 2022 Population and Housing Census Report, Volume 1* [official census report]. https://www.zimstat.co.zw/wp-content/uploads/Demography/Census/2022_PHC_Report_27012023_Final.pdf

Authorship basis: registered authority used as corporate author

Source note: PM-S011 supplies Population by reported mother tongue for Zimbabwe. Registered locator: Table 2.17, printed p. 123. Registered temporal field: 2022. Caveat: Census labels do not resolve every dialect or orthographic choice. Sign Language is a census mother-tongue category, not a complete sign-language-use measure.

PM-S013

Statistics Mauritius. (2022). *2022 Housing and Population Census, Volume II: Demographic and Fertility Characteristics* [official census report]. https://statsmauritius.govmu.org/Documents/Census_and_Surveys/Census2022/HPC_TR_Vol2_Demography_Yr22.pdf

Authorship basis: registered authority used as corporate author

Source note: PM-S013 supplies Resident population by language usually spoken at home for Mauritius. Registered locator: Table D9, printed p. 165, Republic of Mauritius row. Registered temporal field: 2022. Caveat: Chinese languages is an aggregate. Home language is not necessarily L1. The published table excludes Agalega and St Brandon as documented in the report.

PM-S014

Stats NZ. (2023). *Place and ethnic group summaries: Total New Zealand population language comparator* [official census summary table]. <https://tools.summaries.stats.govt.nz/ethnic-group/pacific-peoples-not-elsewhere-classified>

Authorship basis: registered authority used as corporate author

Source note: PM-S014 supplies Persons who reported speaking each language; multiple-response census measure for New Zealand. Registered locator: Table 'Percentage of population that speak each language', Total New Zealand population, 2023 column. Registered temporal field: 2023. Caveat: Multiple responses are permitted and figures use fixed random rounding; rows are not L1 counts and must not be summed. Northern Chinese is a census category, not a single unqualified language target.

Local evidence status: No local witness path or hash is registered; do not describe this source as locally hash-verified.

PM-S016

Central Statistics Office Ireland. (2022). *Census of Population 2022 Summary Results: Education and Irish Language* [official census release]. <https://www.cso.ie/en/releasesandpublications/ep/p-cpsr/censusofpopulation2022-summaryresults/educationandirishlanguage/>

Authorship basis: registered authority used as corporate author

Source note: PM-S016 supplies Population aged 3 and over reporting ability to speak Irish; and daily use outside education for Ireland. Registered locator: Table 6.2 and Figure 6.4 text. Registered temporal field: 2022. Caveat: Ability and daily-use measures are not interchangeable; cells are separated and must not be summed.

Local evidence status: No local witness path or hash is registered; do not describe this source as locally hash-verified.

PM-S020

Central Statistical Agency of Ethiopia (now Ethiopian Statistical Service). (2007). *Population and Housing Census 2007: National Statistical Report* [official census statistical report]. https://ess.gov.et/wp-content/uploads/2007/09/Population-and-Housing-Census-2007-National_Statistical.pdf

Authorship basis: registered authority used as corporate author

Source note: PM-S020 supplies Population by mother tongue used with family members or guardians during childhood for Ethiopia. Registered locator: Table 3.2, printed pp. 91-93, Urban + Rural both sexes. Registered temporal field: 2007. Caveat: The observation is old but remains the latest Ethiopian national census. Several census labels, especially Guragiegna, aggregate distinct languages; counts are not current projections.

PM-S021

BPS-Statistics Indonesia. (2022). *Jumlah Penduduk Berumur 5 Tahun ke Atas Menurut Wilayah, Jenis Kelamin, dan Kemampuan Berbahasa Indonesia, di INDONESIA* [Long Form Population Census 2020 table]. <https://sensus.bps.go.id/topik/tabular/sp2022/196>

Source type and time: Official long-form census table; 2022 release/measure.

Source note: PM-S021 supplies the total population aged five or older reported able to understand spoken Indonesian and produce Indonesian words intelligible to another person: 248,501,794 able and 5,177,554 unable, total 253,679,348. This is oral functional ability, not written literacy, home language, mother tongue, academic reading comfort, or demand for a particular curriculum. The model uses it only as the gross Bahasa Indonesia reach ceiling and applies written-access and non-overlap factors separately.

TH-S001

National Statistical Office Thailand. (2010, September 1). *The 2010 Population and Housing Census (Whole Kingdom)* [official national census report]. https://catalogapi.nso.go.th/api/doc/departement/D10/SD10_04/SD10_04_165_2.pdf

Authorship basis: registered authority used as corporate author

Source note: TH-S001 supplies Population by usual languages spoken at home for Thailand. Registered locator: Table 7, PDF pp. 98-99; census day described on PDF pp. 49-50. Registered temporal field: 2010-09-01. Caveat: The table mixes mutually exclusive Thai-only/Thai-plus-other/other-only strata with named other-language categories that can overlap the Thai-plus-other stratum. English labels contain several typographic or country-name substitutions. Counts are 2010 observations, not current projections; the report warns that independently rounded totals may differ slightly.

ZA-S001

Statistics South Africa. (n.d.). *Census 2022 in Brief* [official national census report]. Retrieved August 25, 2026, from <https://www.statssa.gov.za/publications/Census2022inBrief/Census2022inBriefJune2024.pdf>

Authorship basis: registered authority used as corporate author

Source note: ZA-S001 supplies Population aged 1 year and older by language most often spoken in the household for South Africa. Registered locator: Table 3.8 (1)-(2), printed pp. 26-27; PDF pp. 34-35. Registered temporal field: 2024-06. Caveat: The measure is language most often spoken with other

household members, not mother tongue, proficiency, literacy, or academic-language comfort. The table excludes Not applicable (638,736) and Unspecified (258,967). Census label 'Sign language' does not identify a signed-language variety.

Profile and orthography authorities

S-ACS

U.S. Census Bureau. (n.d.). *Detailed Languages Spoken at Home and Ability to Speak English for the Population 5 Years and Over in the United States: 2017-2021* [official census workbook]. Retrieved August 25, 2026, from <https://www2.census.gov/programs-surveys/demo/tables/language-use/2017/2017-2021-acs-lang-tables-nation.xlsx>

Authorship basis: registered institution used as corporate-author candidate; no personal creator or publication-date field is registered

Source note: S-ACS is profile/orthography authority for the interventions listed here. Exact registered evidence: Nation worksheet supplies the retained point estimates and 90% margins of error; universe is age 5+ language spoken at home. Authority level: primary national statistical authority. The register supplies no publication date; n.d. and the access date are retained rather than inferred.

S-APW

Dilzhe'e Apache Dictionary. (n.d.). *The Pronunciation and Spelling of Dilzhe'e Apache* [community dictionary orthography]. Retrieved August 25, 2026, from <https://dilzhee.western-apache.org/Spelling>

Authorship basis: registered institution used as corporate-author candidate; no personal creator or publication-date field is registered

Source note: S-APW is profile/orthography authority for the interventions listed here. Exact registered evidence: Defines the dictionary spelling and its explicit changes from White Mountain and San Carlos Western Apache practice. Authority level: community language-resource authority. The register supplies no publication date; n.d. and the access date are retained rather than inferred.

S-CH-GU

Kumision i Fino' CHamoru yan i Fina'naguen i Historia yan i Lina'la' i Taotao Tano'. (n.d.). *Utugrafihan CHamoru Guahan* [official orthography]. Retrieved August 25, 2026, from https://kumisionchamoru.guam.gov/wp-content/uploads/2024/07/utugrafihan_chamoru_guahan.pdf

Authorship basis: registered institution used as corporate-author candidate; no personal creator or publication-date field is registered

Source note: S-CH-GU is profile/orthography authority for the interventions listed here. Exact registered evidence: Complete adopted 2024 Guam CHamoru orthography. Authority level: Guam statutory language commission. The register supplies no publication date; n.d. and the access date are retained rather than inferred.

S-CH-MP

Inetnun Kutturan Natibun Marianas / Kkoor Aramasal Marianas. (n.d.). *Revised Chamorro-English Dictionary introduction* [community dictionary documentation]. Retrieved August 25, 2026, from <https://natibunmarianas.org/dictionary-introduction/>

Authorship basis: registered institution used as corporate-author candidate; no personal creator or publication-date field is registered

Source note: S-CH-MP is profile/orthography authority for the interventions listed here. Exact registered evidence: States there are two standard orthographies and that all headwords use the official CNMI orthography adopted in 2010. Authority level: CNMI community language institution. The register supplies no publication date; n.d. and the access date are retained rather than inferred.

S-CHO-MS

Mississippi Band of Choctaw Indians. (n.d.). *Mississippi Band of Choctaw Indians 2023 Tribal Profile* [official tribal publication]. Retrieved August 25, 2026, from https://www.choctaw.org/wp-content/uploads/2024/01/2023-Tribal-Profile_v19-updated.pdf

Authorship basis: registered institution used as corporate-author candidate; no personal creator or publication-date field is registered

Source note: S-CHO-MS is profile/orthography authority for the interventions listed here. Exact registered evidence: Describes the modern Mississippi Choctaw alphabet as a variant of the Byington alphabet and documents tribal language programming. Authority level: tribal government. The register supplies no publication date; n.d. and the access date are retained rather than inferred.

S-CHO-OK

Choctaw Nation of Oklahoma School of Choctaw Language. (n.d.). *Choctaw alphabet* [official tribal-language reference]. Retrieved August 25, 2026, from <https://dictionary.choctawnation.com/alphabet/>

Authorship basis: registered institution used as corporate-author candidate; no personal creator or publication-date field is registered

Source note: S-CHO-OK is profile/orthography authority for the interventions listed here. Exact registered evidence: Defines the Choctaw Nation instructional alphabet. Authority level: tribal government language authority. The register supplies no publication date; n.d. and the access date are retained rather than inferred.

S-DAKOTA-DIO

Dakhota lapi Okhodakichiye. (n.d.). *About Dakhota lapi Okhodakichiye* [community language institution]. Retrieved August 25, 2026, from <https://dakhota.org/about-dio/>

Authorship basis: registered institution used as corporate-author candidate; no personal creator or publication-date field is registered

Source note: S-DAKOTA-DIO is profile/orthography authority for the interventions listed here. Exact registered evidence: Delimits the organization to Eastern/Santee-Sisseton Dakota language work. Authority level: community authority. The register supplies no publication date; n.d. and the access date are retained rather than inferred.

S-DAKOTA-SWC

Sisseton Wahpeton College. (n.d.). *Dakota Dictionary introduction* [tribal-college dictionary documentation]. Retrieved August 25, 2026, from <https://dakotadictionary.hartman-technology.com/uploads/content/Introduction.pdf>

Authorship basis: registered institution used as corporate-author candidate; no personal creator or publication-date field is registered

Source note: S-DAKOTA-SWC is profile/orthography authority for the interventions listed here. Exact registered evidence: Documents the University of Minnesota orthography used at Sisseton Wahpeton College and contrasts another system. Authority level: tribal college. The register supplies no publication date; n.d. and the access date are retained rather than inferred.

S-GARIFUNA-ALPH

Honduras Secretaria de las Culturas las Artes y los Patrimonios. (n.d.). *Alfabeto Garifuna* [official alphabet]. Retrieved August 25, 2026, from <https://arc.secapph.gob.hn/wp-content/uploads/2025/10/Alfabeto-Garifuna.pdf>

Authorship basis: registered institution used as corporate-author candidate; no personal creator or publication-date field is registered

Source note: S-GARIFUNA-ALPH is profile/orthography authority for the interventions listed here. Exact registered evidence: Publishes the alphabet used for the Honduras educational profile. Authority level: national government authority. The register supplies no publication date; n.d. and the access date are retained rather than inferred.

S-GARIFUNA-HN

Honduras Secretaria de las Culturas las Artes y los Patrimonios. (n.d.). *Unidad de Educacion Plurilingue y Multicultural* [official government education page]. Retrieved August 25, 2026, from <https://arc.secaphh.gob.hn/direccion-general-de-las-culturas-y-los-patrimonios/unidad-de-educacion-plurilingue-y-multicultural>

Authorship basis: registered institution used as corporate-author candidate; no personal creator or publication-date field is registered

Source note: S-GARIFUNA-HN is profile/orthography authority for the interventions listed here. Exact registered evidence: Documents government production of Garifuna educational texts and links the official alphabet. Authority level: national government authority. The register supplies no publication date; n.d. and the access date are retained rather than inferred.

S-HOPI

Hopi Dictionary Project. (n.d.). *Learn Hopi / Third Mesa dictionary materials* [institutional teaching and dictionary resource]. Retrieved August 25, 2026, from <https://hopidictionary.com/wp-content/uploads/2024/10/learnhopi.pdf>

Authorship basis: registered institution used as corporate-author candidate; no personal creator or publication-date field is registered

Source note: S-HOPI is profile/orthography authority for the interventions listed here. Exact registered evidence: Provides a reproducible Third Mesa-oriented Hopi writing and teaching profile. Authority level: community and University of Arizona project. The register supplies no publication date; n.d. and the access date are retained rather than inferred.

S-IANA

Internet Assigned Numbers Authority. (n.d.). *Language Subtag Registry* [standards registry]. Retrieved August 25, 2026, from <https://www.iana.org/assignments/language-subtag-registry/language-subtag-registry>

Authorship basis: registered institution used as corporate-author candidate; no personal creator or publication-date field is registered

Source note: S-IANA is profile/orthography authority for the interventions listed here. Exact registered evidence: Validates emitted language/script/region components and identifies ik as a macrolanguage with esi and esk members. Authority level: primary international registry. The register supplies no publication date; n.d. and the access date are retained rather than inferred.

S-INUPIAQ

Alaska Native Language Archive / University of Alaska Fairbanks. (n.d.). *Guide to the Inupiaq Language Collection* [public-university archive guide]. Retrieved August 25, 2026, from <https://www.uaf.edu/anla/collections/inupiaq/>

Authorship basis: registered institution used as corporate-author candidate; no personal creator or publication-date field is registered

Source note: S-INUPIAQ is profile/orthography authority for the interventions listed here. Exact registered evidence: Identifies two major groups and four dialect types and records an accepted 1970s orthography. Authority level: public university. The register supplies no publication date; n.d. and the access date are retained rather than inferred.

S-INUPIAQ-NS

University of Alaska Fairbanks Documenting Alaskan and Neighboring Languages. (n.d.). *North Slope Inupiaq lexicon* [public-university lexicon record]. Retrieved August 25, 2026, from <https://www.uaf.edu/danl/project-updates/edna-maclean/>

Authorship basis: registered institution used as corporate-author candidate; no personal creator or publication-date field is registered

Source note: S-INUPIAQ-NS is profile/orthography authority for the interventions listed here. Exact registered evidence: Identifies the completed lexicon as the definitive dictionary for North Slope Inupiaq. Authority level: public university. The register supplies no publication date; n.d. and the access date are retained rather than inferred.

S-LKT

Lakota Language Consortium. (n.d.). *Lakota Stories and Lakota Level 1 textbook* [institutional teaching and reader resources]. Retrieved August 25, 2026, from <https://lakhota.org/product/lakota-stories-collected-by-ella-deloria/>

Authorship basis: registered institution used as corporate-author candidate; no personal creator or publication-date field is registered

Source note: S-LKT is profile/orthography authority for the interventions listed here. Exact registered evidence: Defines a consistent contemporary phonemic orthography used in the reader and a pedagogically consistent textbook series. Authority level: language-teaching institution. The register supplies no publication date; n.d. and the access date are retained rather than inferred.

S-ODAWA-DICT

Algonquian Dictionaries Project. (n.d.). *Nishnaabemwin Odawa and Eastern Ojibwe online dictionary* [institutional dictionary]. Retrieved August 25, 2026, from <https://dictionary.nishnaabemwin.atlasling.ca/>

Authorship basis: registered institution used as corporate-author candidate; no personal creator or publication-date field is registered

Source note: S-ODAWA-DICT is profile/orthography authority for the interventions listed here. Exact registered evidence: Identifies an exact Odawa/Eastern Ojibwe dictionary data edition suitable as a named production profile. Authority level: community and academic dictionary project. The register supplies no publication date; n.d. and the access date are retained rather than inferred.

S-ODAWA-TRIBAL

Adawe Cultural Center / Ottawa Tribe of Oklahoma. (n.d.). *Language resources* [official tribal cultural resource]. Retrieved August 25, 2026, from <https://www.heritage.ottawatribe.gov/language>

Authorship basis: registered institution used as corporate-author candidate; no personal creator or publication-date field is registered

Source note: S-ODAWA-TRIBAL is profile/orthography authority for the interventions listed here. Exact registered evidence: Corroborates tribal use and recommendation of Double-Vowel language resources. Authority level: tribal cultural institution. The register supplies no publication date; n.d. and the access date are retained rather than inferred.

S-OJ

University of Minnesota Ojibwe People's Dictionary. (n.d.). *About the Ojibwe language* [institutional dictionary documentation]. Retrieved August 25, 2026, from <https://ojibwe.lib.umn.edu/about-ojibwe-language>

Authorship basis: registered institution used as corporate-author candidate; no personal creator or publication-date field is registered

Source note: S-OJ is profile/orthography authority for the interventions listed here. Exact registered evidence: Names the Central Southwestern variety and states that the dictionary uses the Double-Vowel system. Authority level: public-university and community consortium. The register supplies no publication date; n.d. and the access date are retained rather than inferred.

S-PAP-AW

Government of Aruba. (n.d.). *First Bible in Papiamentu in Aruba's official spelling* [official government language page]. Retrieved August 25, 2026, from <https://www.gobierno.aw/en/first-bible-in-papiamentu-in-arubas-official-spelling>

Authorship basis: registered institution used as corporate-author candidate; no personal creator or publication-date field is registered

Source note: S-PAP-AW is profile/orthography authority for the interventions listed here. Exact registered evidence: Explicitly distinguishes Aruba's official spelling from Curaçao's existing Papiamentu edition. Authority level: territorial government authority. The register supplies no publication date; n.d. and the access date are retained rather than inferred.

S-PAP-CW

Government of Curaçao. (n.d.). *Landsbesluit schrijfwijze Papiamentu en Nederlands* [official spelling decree]. Retrieved August 25, 2026, from <https://gobiernu.cw/wp-content/uploads/2025/12/196-GT.-Lb-schrijfwijze-Papiamentu-en-Nederlands.pdf>

Authorship basis: registered institution used as corporate-author candidate; no personal creator or publication-date field is registered

Source note: S-PAP-CW is profile/orthography authority for the interventions listed here. Exact registered evidence: Establishes the official Papiamentu spelling and wordlist used in Curaçao; the same phonological tradition is the statutory Bonaire baseline. Authority level: territorial government

authority. The register supplies no publication date; n.d. and the access date are retained rather than inferred.

S-PAP-FACT

United Kingdom Foreign Commonwealth and Development Office. (n.d.). *Kingdom of the Netherlands Toponymic Factfile* [official government factfile]. Retrieved August 25, 2026, from https://assets.publishing.service.gov.uk/media/68419e2efa0289a17a3e879c/Kingdom_of_the_Netherlands_Toponymic_Factfile.pdf

Authorship basis: registered institution used as corporate-author candidate; no personal creator or publication-date field is registered

Source note: S-PAP-FACT is profile/orthography authority for the interventions listed here. Exact registered evidence: Distinguishes Aruba Papiamentu from Curaçao/Bonaire Papiamentu usage. Authority level: government institutional source. The register supplies no publication date; n.d. and the access date are retained rather than inferred.

S-PDC

Deitsh Books. (n.d.). *Pennsylvania German spelling-system resources* [community publishing resource]. Retrieved August 25, 2026, from <https://www.deitshbooks.com/tools/books/grammer/>

Authorship basis: registered institution used as corporate-author candidate; no personal creator or publication-date field is registered

Source note: S-PDC is profile/orthography authority for the interventions listed here. Exact registered evidence: States that two written spelling systems exist and identifies the English-oriented system used in named books. Authority level: community language publisher. The register supplies no publication date; n.d. and the access date are retained rather than inferred.

S-YIVO

YIVO Institute for Jewish Research. (n.d.). *Basic Facts about Yiddish* [institutional language reference]. Retrieved August 25, 2026, from https://www.yivo.org/cimages/basic_facts_about_yiddish_2014.pdf

Authorship basis: registered institution used as corporate-author candidate; no personal creator or publication-date field is registered

Source note: S-YIVO is profile/orthography authority for the interventions listed here. Exact registered evidence: Section on Standard Yiddish defines the YIVO literary standard while delimiting it from any

single spoken dialect. Authority level: major cultural and scholarly institution. The register supplies no publication date; n.d. and the access date are retained rather than inferred.